

Chapter | 9 | Technical Research



>>> Chapter Objectives

In this chapter, students will

- Learn how to focus a research project
- Learn how to identify sources that will help them answer their research questions
- Learn how to find and use published research
- Learn how to conduct primary research
- Learn how to use information from sources correctly
- Learn how to present their research findings to others
- Read and analyze a sample research report

Tanya Grant, who works in marketing at M-Global's Atlanta office, has just been given an important task. The company president, Jim McDuff, wants her to examine the feasibility of the company's switching to hybrid electric-powered cars in their American offices. McDuff believes the company's success in using these vehicles in their Asian offices is so significant that the company should consider using them in the U.S. offices. Success with hybrid electric cars could help address serious air quality issues and offset escalating petroleum prices. Also, Jim hopes his firm will become a major player in refining charging-station technology and fostering its use. In short, this move could serve as a public relations effort, a budgetary control, and a marketing tool for M-Global products.

Jim has told Tanya she should write a report investigating the advantages and disadvantages of hybrid electric cars. At an upcoming meeting, upper-level management will review the report. She should study the impending tax and other legislation affecting fuel-efficient vehicles. In addition, she should investigate economic feasibility of hybrid vehicles. Some of her report will look at start-up costs for switching

the vehicle fleet, employee needs, and other in-house matters.

In your classes, your teachers often assign you a general topic or question to research and write about. They may expect you to find your own narrow focus for your response, and they probably ask you to depend primarily on print sources available in the library or on the Internet. Research papers are an important way for teachers to see how you learn about and analyze an issue.

Research does not end with the last college term paper. In the workplace, you may write articles that are much like the papers you have written in school for professional journals, but much of the research you write about as part of your job will serve a different purpose than your writing in your classes. Figure 9-1 shows some of the differences between research in school and research in the workplace. Workplace research aims to explain an issue or help the reader solve a problem. Research may be assigned by managers, or writers may decide to conduct research and write the results as a report or presentation.

Even though your workplace research serves a different purpose and may be published in a different

Features	Writing prompt	Purpose	Audience	Sources	Publication format
Academic writing	General topic assigned by the teacher	Communicating what the student knows about the topic, to earn a high grade	The teacher who assigned the project	Secondary sources, for the most part	Academic papers Presentations and posters at academic conferences
Workplace writing	Specific workplace situation, question, or problem raised by the writer or by a supervisor	Providing information needed to answer a question or make a decision	Often several people with differing professional backgrounds	Secondary sources serve as foundation for primary research	Reports Proposals Workplace presentations Presentations and posters at professional conferences

■ **Figure 9-1** ■ Features of academic and workplace research projects

format than academic papers, it starts with the same review of published articles that you have learned to do in your college writing classes. In fact, your career will often require you to gather technical information from libraries, the Internet, and other sources. Such on-the-job research produces documents as diverse as reports, proposals, conference presentations, published papers, newspaper articles, Web sites, or essays in company magazines. Your professional reputation may depend on your ability to locate information, evaluate it, and use it effectively in everyday research tasks.

This chapter takes you through the research process practiced on the job. Specifically, the chapter has five main sections:

1. Getting started
2. Reviewing published research
3. Conducting primary research
4. Using borrowed information correctly
5. Reporting your research

A common thread throughout the chapter is the M-Global case study of Tanya Grant's project for Jim McDuff. We'll observe Tanya as she gathers research material. Like Tanya, in your career you will have to apply the research process on the job. It is one thing to read about doing research; it is quite another to dive into your project and work directly with the books, periodicals, electronic databases, Web sites, and other resources in the library and on the Internet. You should seek firsthand research experience as soon as you can.

Finally, remember that the best research writing smoothly merges the writer's ideas with supporting data. Such writing should (1) impress the reader with its clarity and simplicity and (2) avoid sounding like a strung-together series of quotations. These two goals present a challenge in research writing.

>>> Getting Started

In Chapter 2, you learned about the three phases of any writing project: planning, drafting, and revising. Research can occur in the planning stage, right before you complete an outline, and it can also occur again and again throughout the project. Before starting your research, ask yourself questions like the following to give direction for your work:

- What questions must be answered during the research phase?
- What *secondary sources*, including print, multimedia, and electronic sources, are most useful?
- What is the nature and extent of information that is needed? Should it be scholarly or popular? Current or historical?
- What are the best strategies and research tools for locating information?
- What *primary sources*, including interviews, surveys, field observations, and usability tests, will provide useful information?
- What are the best criteria for critically evaluating information for reliability, validity, accuracy, timeliness, or point of view or bias?
- What format must be used to document material borrowed from sources, and what copyright permissions must be acquired to use the information?

The following outline shows how Tanya answers the questions previously noted as she begins her research:

1. Main question: Should M-Global switch to hybrid electric vehicles?
2. Main types of information needed:
 - What are the advantages and disadvantages of this technology?
 - What is the research saying about the outlook of hybrid electric vehicles?
 - What tax or other legislation is pending at both state and federal levels?
 - Who are the current and potential consumers, and what do they think about hybrid electric cars?
 - What are the real costs in maintaining a fleet?
 - What could M-Global gain by the switch? What would it lose?
 - What impact would such a switch have on the competition, potential clients, or employees?
3. Possible sources:
 - Memos, reports, and other M-Global documents related to the use of hybrid cars in the organization's Asian offices
 - Directories (of periodicals, newsletters, newspapers, electronic journals, organizations)
 - Journal and newspaper articles found in indexes, abstracts, and electronic databases
 - Bibliographies and literature reviews
 - Government documents
 - Books
 - Web sites
 - Surveys
 - Interviews
4. Format for documentation: Tanya submits a short report to Jim McDuff, documenting her research using the *Publication Manual of the American Psychological Association's* (APA) system for citing borrowed information (the same format used by M-Global engineers and scientists in their research reports).

>>> Reviewing Published Research

All research draws in some way on research that has been conducted and published by others, so most researchers begin by reading these *secondary sources*. Secondary sources can be defined as follows:

Secondary sources: Information about a topic that has been shared through print, recorded media, or presentations. Secondary sources provide researchers and readers with the background information they need by establishing the professional and intellectual context for an issue or problem.

Even an internal report like Tanya Grant's needs to examine materials that have already been published about the topic. It may be that someone has already studied the issue and provided the data you need, or someone has recommended a solution. Even if the information that you find is specific to another organization, a published study may offer you a useful methodology to collect information that will address the situation in your own organization. With her basic plan in mind, Tanya can begin her work. Her first decision is where to start her research. Locations for secondary sources include the following:

- The public library
- Her corporate library
- The university library
- The World Wide Web

She eliminates the public library as too general, and the M-Global corporate library contains only a copy of the report about the Asian offices' use of hybrid vehicles. Jim McDuff has already given her a copy of this report. The remaining options are the Web and the local university libraries. Tanya must use both to do a thorough job of research. She uses the Web regularly for finding business information, news, entertainment, and discussion on just about anything. Although she is not a student at the university, Tanya has a borrower's card that gives her access to one library's databases and collections. She knows about the university's extensive print collections in science and technology, its Web-based online catalog, and its extensive collection of electronic databases. In addition, the staff in the reference department and the interlibrary loan office will help her identify and track down sources. The two, the Web and the library, work amazingly well together—they intersect, overlap, and complement one another, and each contributes unique sources, provided you know the basic research techniques and tools. A good researcher uses both the Web and the library to take advantage of the strengths and overcome the weaknesses of both.

A planning trip to the library can save you hours of time searching the Web in unfamiliar subject areas. However, a few hours' preparation online—for example, searching a library's Web-based catalog—can make your trip to the library more productive.

Tanya schedules two days the first week and one the next week to work in the library, and she begins her research from her computer in the office. The next sections introduce some of the strategies, tools, and basic concepts you need for searching library online catalogs, searching in the library, and searching on the Web.

Searching Online Catalogs

Both the library and the Internet can seem like intimidating places when you first start a project. Once you learn a few basics, however, you will become comfortable and even confident about using these



resources. This section includes information on using online library catalogs to locate books, journals, and other resources. We look at the basics as well as more advanced techniques you need for searching not only library catalogs, but also most Web search engines and electronic databases.

Books and other printed sources provide well-supported and tested information about a topic, but by definition, the information is often dated. Even a book just published has information that is one to two years old, given the time it takes to put a book-length manuscript into print. Keep this limitation in mind as you search.

The library's catalog is a road map to its collection of books, periodicals, and other material; it is an alphabetical list by author, title, and subject. These days, the majority of college and university libraries offer sophisticated online catalogs that can be searched at the library or remotely from home or office. The online catalog often has additional features such as keyword and Boolean searching and information about whether a book is available or checked out. The rules for searching online catalogs vary depending on the computer program used by the library. The online catalog's Help screen is the best guide to search techniques. Following are some general strategies for effective searching.

Author or Title Search

If you know specific authors or titles of potentially useful books, conduct an author or title search to locate the library's call numbers. Study the catalog entry, especially the subject heading; similar books can be found if you search by subject using these terms. Often online catalogs feature automatic links to these terms.

Subject Search

Once you know the subject headings assigned to books or resources on your topic, searching by subject can be very efficient. Libraries select the subject headings from *The Library of Congress Subject Headings*. Unlike the Web, a library's catalog has subject terms that are controlled and very specific in order to bring all the material together. Knowing exactly which subject words to use can be a matter of trial and error, but once found, these headings can serve as powerful tools to gather information on your topic.

Keyword Search

This strategy is probably your best choice because it allows you to scan through all the fields in a book's library record—author field, title, subject headings, dates—to locate books that match your request. Pay close attention to the catalog's rules for keyword searches; you can often improve your results by limiting searches to particular fields. Figure 9–2 shows a typical keyword search.

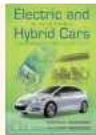
Tip: One of the easiest ways to find material on your topic is to switch between keyword and subject searching. For example, begin with a keyword search, selecting the books that match your request and the subject headings used to describe those books. Then do a subject search.

Author [Anderson, Curtis D. \(Curtis Darrel\), 1947-](#)

Title [Electric and hybrid cars : a history / Curtis D. Anderson and Judy Anderson.](#)

Published Jefferson, N.C. : McFarland, c2010.

Edition 2nd ed.



[Reviews & More](#)

LOCATION	CALL #	STATUS
MWSU Stacks - Upstairs	TL220 .A53 2010	AVAILABLE


[Send Call Number via Text](#)


Description x, 257 p. : ill ; 26 cm.

Bibliography Includes bibliographical references and index.

Summary "This illustrated history of **electric** and **hybrid** vehicles covers the companies that produced various models; the politics that have surrounded them; the environmental aspects of **electric** and **hybrid** vehicles versus internal combustion engines; efforts to overcome technological challenges associated with **electric** vehicles; marketing strategies through the decades; and public attitudes towards these vehicles throughout their existence"--Provided by publisher.

Contents Introduction : The birth of the automobile industry -- The evolution of the **electric** vehicle -- Politics -- Environment -- Technology -- Marketing -- Conclusion -- Appendix : Hybrids defined.

LC SUBJECT [Electric automobiles.](#)
[Hybrid electric cars.](#)

ISBN 9780786433018 (softcover : alk. paper)
0786433019 (softcover : alk. paper)

Standard # NZ1 13287211
CDX 10371845
AU@ 000045360163
NLGGC 327216263

OCLC # 424560321

■ **Figure 9-2** ■ Results of a typical keyword search

Source: Missouri Western State University Library Catalog.

Advanced Search Techniques

A library catalog may also include advanced strategies such as Boolean searching, positional operators, and truncation. Many times you may not be aware that you are using these tools because they are built into the catalog's search functions. However, learning to use these techniques is important because they are used in library catalogs, most periodical databases, and Web search engines. Look for the Help screens in your catalog that describe the advanced search options, and practice using them whenever possible; they can save you time and produce excellent results. Following is a brief description of some of the most common search options:

- A **Boolean search** outlines the relationship of words and phrases using simple AND, OR, NOT statements (Figure 9-3).
- **Positional operators** stipulate the relative location of each term within the record. For example, you can often specify that terms must be adjacent or within a certain number of words.
- **Truncation** allows for variant spelling or plurals. For example, in some catalogs entering *wom*n* retrieves records including either of the words *woman* and *women*.

Searching other library catalogs can sometimes be as simple as selecting a link from your library's Web site to a library consortium or union catalog of university

Boolean Searching**AND:** Example: cars AND SUVs

Locates only those records where both terms are present.

Use this to narrow your search and reduce the number of matches.

OR Example: cars OR automobiles OR SUVs

Locates records in which any one of these terms can appear.

Use this to broaden or enlarge your search.

NOT: Example: SUVs NOT trucks

Eliminates records containing the excluded term.

Use this sparingly to narrow your search.

■ **Figure 9-3** ■ Boolean search examples

Tip: Cite your sources as you go. Keep close track of what you find and where you find it so that you don't waste your time searching for books on the shelves of your library when they are actually located elsewhere. Consult the reference department of your library to learn about your library's interlibrary loan service or borrowing privileges at other libraries.

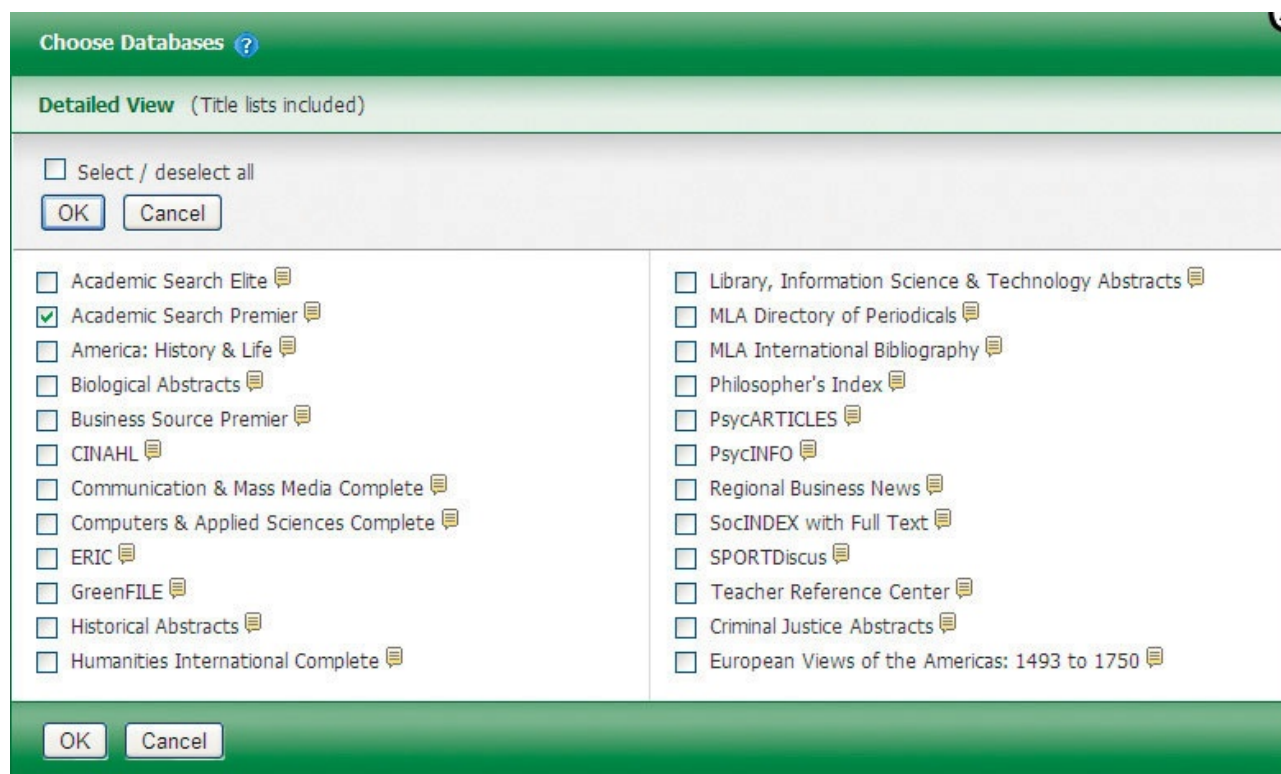
and college libraries within your region. Searching libraries close to home has the advantage of easier access to their collection, whether you visit in person or gain access through your library's interlibrary loan service. If you want to see "what's out there" in larger or more specialized libraries, try searching for library

catalogs on the Web or ask your reference librarian if your library provides access to Worldcat (<http://www.worldcat.org>) or the Online Computer Library Center (OCLC) site (<http://www.oclc.org/us/en/global/default.htm>). Two Web directories of library catalogs that have been around for some time are Libcat: A Guide to Library Resources on the Internet—Libraries in the United States (<http://www.librarysites.info>) and Libweb: Library Servers (<http://lists.webjunction.org/libweb/>). (Keep in mind, however, that because of the transitory nature of the Web, they may have disappeared since this book was printed.)

When you use the Web for searching catalogs, remember to evaluate what you find as you search. Begin thinking critically as soon as you start, and work to keep this perspective throughout your research. For example, when looking at a reference to a book, a journal, an article, or a Web site, ask yourself the following questions:

- What are the author's academic or professional qualifications?
- Who is the publisher, and what is its reputation?
- What are the scope and content of the work?
- How does this information fit in with what you know about this topic?
- What are the trends in information on this topic, and how does this book, article, journal, or Web site fit in?
- How current is this information?

Finally, use your library's online catalog to find out what electronic databases are available to you. You may be able to search important research databases with access to periodicals, newspapers, encyclopedias, dictionaries, directories, statistics, and other



■ **Figure 9-4** ■ Research databases available on one library's Web site

Source: Missouri Western State University Library Online.

reference sources (Figure 9-4). Many of these databases can be searched remotely from your home or office, but some are restricted to in-library searching only. Policies governing who can search, from where, passwords, and whether searching is fee-based or free vary widely depending on the contracts between the library and the database vendor. The next section of this chapter covers electronic databases in more depth; for now, keep in mind that the quality of information you retrieve from research databases is usually superior to the material you may locate searching the vast World Wide Web. In addition, online catalogs may provide links to recommended high-quality Web sites you might not otherwise locate. Explore your online options and discuss your needs with the reference staff at your library.

Tanya Grant's M-Global Project

Tanya Grant, for example, rightly thinks that books will not be her main source of information about hybrid electric cars because the topic has developed relatively recently, but she at least wants to see what range of sources the catalog offers. Tanya does a keyword search and determines that the correct subject heading is "hybrid electric vehicles." Her own library's holdings are somewhat limited, but one item is worth reviewing. She decides to search the online catalog from another local university with an automotive engineering school, where she finds a better selection of books, and because her library card gives her borrowing privileges



newphotoservice/Shutterstock

at all state university system libraries, she decides to take a trip to this library. Her final search is in OCLC and Webcat, databases that lead her to a few more noteworthy titles that she will borrow through her library's interlibrary loan department.

Searching in the Library

At some point during your search for secondary resources, you should visit an academic library. The library's services and collections of books, journals, electronic databases, microforms, and reference materials, although complex, support your research and help you locate information. Fortunately, academic and research libraries are organized along similar principles,

and the skills you gain from using one library can generally be used at other libraries as well. This section highlights some of the services and resources you can expect to find as you conduct your research in the library.

Library Resources

This section includes information on the following resources: books; periodicals; newspapers; company directories; and dictionaries, encyclopedias, and other general references.

>> Resource 1: Books

As previously discussed, the library catalogs these days are generally automated. Once you locate the exact book for which you are searching, browse through the books located beside this title. You are likely to find other useful and related material. You may also find that the book you need is available as an e-book. Your library's Web site will have information about how to check out, download, and open e-books. Ask for assistance at the reference or circulation desk if you cannot locate the books on your topic or if you don't have access to an e-book that you need.

>> Resource 2: Periodicals

Periodicals are publications that are issued on a regular basis, usually weekly, monthly, or quarterly. The term encompasses

- Popular magazines that take commercial advertising, such as *Time*, *Science*, and *National Geographic*
- Professional and scholarly journals such as *IEEE Transactions on Professional Communication* and *Technical Communication Quarterly*

Your key tool for locating information within periodicals is an electronic database of periodical indexes or abstracts. By looking up your subject in the index, you can find articles that provide the information you need. Some databases, like Academic Search Premier

include popular periodicals. Others, like the Engineering Index, deal with a broad range of technical information. Still others, like Mechanical Engineering Abstracts, focus on periodicals, books, Web sites, and papers in specialized technical fields. The periodicals covered in the index or abstract are listed in the volumes or in the online information screen, along with the inclusive dates of the issues indexed.

Sometimes an abstract, or a summary of the article, is all that is available to you; it provides a brief description of articles so that you can decide whether the entire article is worth finding. Abstracts are especially useful when the article being summarized is not available in your library. The abstract can help you decide whether to (1) visit another library, (2) order the article through the interlibrary loan service, or (3) disregard the article altogether.

Increasingly, electronic databases provide full-text copies of the periodical articles. Some libraries permit you to search these databases from your home or office, whereas other libraries, because of the license requirements of the database vendors, permit searching within the library only. Still other libraries provide professional search services where, for a fee, the research staff conducts the search for you.

The rules for searching electronic databases vary widely. Each database has unique features and searching requirements. You must invest time and energy to learn these rules to take full advantage of the information the database offers. Start your search by reading the Help screens and the instructional materials about the database or any support materials that the library provides. You will save yourself time and improve your search results if you understand the basic search strategies and have a grasp of the scope of the database. At a minimum, make sure that you know the rules for printing, e-mailing, or saving to disk the results of your search before you get too far into your research.

Most of the electronic databases have search strategies similar to what you may have encountered when searching the online catalog for books, and they are likely to include subject searching, keyword searching, advanced search techniques using Boolean and positional operators, truncation options, and language- and date-limiting options. Also common are options to limit searches to scholarly or peer-reviewed journals, or to full-text journals. The more you practice, the better your searching and the more precise your results.

Tip: E-mailing results from a search in an electronic database is an efficient and accurate way to collect the information you need to document your research and build your work-cited page.

There are hundreds of electronic databases and print indexes or abstracts available. Many libraries provide guides to these resources. Ask the reference staff to help you locate the most appropriate ones for your topic. Following is a list of a few of the well-known titles available in print or electronically:

- Academic Search Premier
- Applied Science and Technology Abstracts (print title: *Applied Science and Technology Index*)
- ABI/Inform Complete at ProQuest
- BIOSIS: Biological Abstracts

- CSA: Cambridge Scientific Abstracts
- CAS: Chemical Abstracts
- Computer Abstracts International Database
- Current Contents
- EI: Engineering Information
- General Science Abstracts (print index: *General Science Index*)
- GPO Monthly Catalog (index to government documents)
- Inspec, the database of the Institution of Engineering and Technology
- Lexis-Nexis Academic Universe
- PsycINFO (print title: *Psychological Abstracts*)
- Science Citation Index, Social Science Citation Index, Arts & Humanities Citation Index (online through the Web of Science)

Tanya Grant's M-Global Project

Tanya decides to consult a few of the electronic databases recommended by the reference librarian.

1. She conducts a search using GreenFILE, which the library subscribed to electronically. The scope and content of the abstract are exactly what she wants because they target the technological and engineering aspect of hybrid electric vehicles. Because the database is new to her, she spends time learning how to conduct a search and save her results. She limits her search to scholarly and peer-reviewed articles from the last few years. The search not only retrieves useful articles but also provides links to six high-quality Web sites. Scanning the results, she selects the most promising articles and Web site and e-mails a copy to herself and prints a copy of the list to use for locating the periodicals in the library. Figure 9–5 shows Tanya's primary search.

2. Next she consults ABI/Inform Global, an online database that covers business and management trade journals produced by ProQuest. She's interested in looking at business viewpoints on hybrid vehicles. Her search produces 72 items published since 2008, many of which have full-text copies of the article available for her to read immediately. After sampling a few articles, she flags those she wants and e-mails them to herself. Tanya decides to redo her search and narrow it to peer-reviewed articles only. The nine articles she retrieves in her second search have undergone review and evaluation by experts in the field prior to publishing. These articles will be particularly noteworthy.

3. Finally, Tanya consults Academic Search Premier, a comprehensive, general-purpose database. It covers almost 4,000 periodicals, 2,300 of which are scholarly. Again, she is able to narrow her search to peer-reviewed articles and locates some very current and useful information. One of the full-text articles refers to an organization she wants to investigate further, the Partnership for a New Generation of Vehicles. Figure 9–6 shows Tanya's search.

The screenshot displays the EBSCO GreenFILE search interface. At the top, the search criteria are: "hybrid" AND "electric" AND "vehicles". The search results are displayed in a list format, showing two results. The first result is titled "Effect of hybrid system battery performance on determining CO₂ emissions of hybrid electric vehicles in real-world conditions." and the second result is titled "Integration of plug in hybrid electric vehicles in a regional wind thermal power system.".

Search Options

- Search modes:**
 - ☒ Boolean/Phrase
 - ☐ Find all my search terms
 - ☐ Find any of my search terms
 - ☐ SmartText Searching [Hint](#)
- Apply related words:** ☐
- also search within full text articles:** ☐

Limit your results

- Full Text:** ☐
- Scholarly (Peer Reviewed) Journals:** ☒
- Publication Type:**
 - All
 - Academic Journal
 - Book/Monograph
 - Essay
- References Available:** ☐
- Publication:**
- Publication Date:**
 - Month: Year: 2008 to
 - Month: Year: 2011
- Document Type:**
 - All
 - Abstract
 - Article
 - Bibliography

Search Results

110 Results for...
Doolean/Phrase: hybrid and electric and vehicles

Limiters

- ☒ Scholarly (Peer Reviewed) Journals
- ☒ Publication Date: 2008 to 2011

Refine your results

- ☐ Full Text
- ☐ References Available
- ☒ Scholarly (Peer Reviewed) Journals

Results:

- Effect of hybrid system battery performance on determining CO₂ emissions of hybrid electric vehicles in real-world conditions.**
By: Alvarez, Robert; Schlenger, Peter; Weidenmann, Martin. Energy Policy, Nov2010, Vol. 38 Issue 11, p6919-6925, 7p; DOI: 10.1016/j.enpol.2010.07.008
Subjects: HYBRID electric vehicles; CARBON dioxide; EMISSIONS; ELECTRIC vehicles; DYNAMOMETER
Database: GreenFILE
Add to folder | Relevancy: ***** | Cited References: (1)
Check Article Linker for Text
Notes: This title is not held locally
- Integration of plug in hybrid electric vehicles in a regional wind thermal power system.**
By: Göransson, Lisa; Karlsson, Sten; Johnsson, Filip. Energy Policy, Oct2010, Vol. 38 Issue 10, p5482-5492, 11p; DOI: 10.1016/j.enpol.2010.04.001
Subjects: PLUG-in hybrid electric vehicles; HYBRID electric vehicles; WIND power; EMISSIONS; ATMOSPHERIC carbon dioxide
Database: GreenFILE
Add to folder | Relevancy: *****

■ **Figure 9-5** ■ Results of a search conducted in GreenFILE through a library's Web site

Source: GreenFILE through Missouri Western State University Library.

Two particularly noteworthy newspapers are the *New York Times* and the *Wall Street Journal*. These well-respected newspapers have a long tradition of high-quality journalism. Both titles are thoroughly indexed, and many libraries either provide access through an electronic database or keep print indexes and back issues in microfilm or microfiche.

Many other regional, national, and international newspapers have established Web sites at which you can frequently locate the archives or find additional information not available in the print version. Two significant electronic databases for newspapers are Lexis-Nexis' Academic Universe, a full-text index to some 5,000 publications, including newspapers, wire services, legal news, and government publications and ProQuest Newspapers, an index to five major newspapers. Check in your library's online catalog to see if it provides additional links to some of the Web-based news services.

Tanya Grant's M-Global Project

Tanya decides to see what kind of newspaper coverage hybrid electric cars are receiving and try to uncover some of the tax legislation being proposed by each state. Her first search in the Lexis-Nexis Academic Universe locates 964 newspaper articles—some written just the previous week. With her second search, Tanya adds the concept “tax” and uncovers 135 articles from major newspapers from around the world describing various tax legislation efforts under way. She is able to search articles in specific newspapers and finds seven articles in the *New York Times* and one article in the *Mobile [Alabama] Register*. These articles serve as a starting point for studying the complex tax legislation being proposed by various state legislatures. Figure 9–7 shows Tanya's search results.

>> Resource 4: Company Directories

Often your research needs may require that you find detailed information about specific firms. For example, you could be completing research about a company that may hire you, or you may seek information about companies that compete with your own. Today, you can find many databases of company information online. In addition, most companies now produce sophisticated Web sites about their services and products. Although not without bias, these can be an excellent source of information. The following is a small sample of some useful directories that are available; ask the reference librarian to recommend others and to assist you in using the online versions of these and other directories.

Compact D/SEC

Corp Tech Directory of Technology Companies

D & B Million Dollar Directory

Mergent Online

Standard & Poor's Register of Corporations, Directors, and Executives

Ward's Business Directory of U.S. Private and Public Companies

Who's Who in Science and Engineering

LexisNexis® Academic

Home

General Searching

» Easy Search™

» Power Search

Tip: Click the headings below to view links to specialized and other vi

Easy Search™

Use of this service is subject to Terms and Conditions

Search the News

Look up a Legal Case

Get Company Info

By Source Type

By Citation like 353 U.S. 537

By Name like Microsoft

Or by Parties like *Mapp v. Ohio*

Or by Ticker like *MSFT*

Or by Source Title

Or by Topic like *Equal Opportunity*

Start typing a title like *New York Times*

» Try also All News search & Sources

» Try also Cases, Landmark Cases

» Try also Company Dossier search &

LexisNexis® Academic

Results List | Edit Search | New Search | Home

Result Groups:

News (135)

McClatchy-Tribune Business News (8)

The New York Times (7)

The Toronto Star (5)

Chicago Sun-Times (4)

Korea Herald (4)

San Antonio Express-News (4)

Automotive News (3)

The Calgary Sun (3)

Edmonton Journal (3)

Financial Post Investing (3)

The Globe and Mail (Canada) (3)

Kamloops Daily News (British Columbia) (3)

USA Today (3)

The Vancouver Sun (3)

The Business Times Singapore (2)

Chattanooga Times Free Press (2)

Chicago Daily Herald (2)

The Economic Times (2)

The Gazette (Montreal) (2)

The Hamilton Spectator (2)

McClatchy-Tribune News Service (2)

Modesto Bee (2)

Pittsburgh Post-Gazette (2)

St. Louis Post-Dispatch (2)

The Standard (St. Catharines) (2)

The Sunday Times (London) (2)

The Sydney Morning Herald (Australia) (2)

The Age (Melbourne, Australia) (1)

Alberta Valley Times (British Columbia) (1)

The Australian (1)

Sort: Relevance

1-25 of 135

View Tagged

Duplicate Options: off

What's this?

Selected Group: Newspapers (View All Results)

Results:

1. How the road tax on Lexus RX400h is derived
2. The Straits Times (Singapore), January 30, 2007 Tuesday, 344 words, Naleeza Ibrahim (My) Deputy Director, Media Relations Land Transport Authority IRS to phase out tax break for hybrid vehicles: Honda, Toyota benefiting (All but Vancouver edition headline.); IRS to phase out tax break for hybrid vehicles: Honda, Toyota to benefit (Vancouver edition headline.)
3. National Post (Canada), May 22, 2002 Wednesday National Edition, Financial Post: World; Pg. FP20, 353 words, Alison Fitzgerald
4. Ford Escape hybrid qualifies for IRS break
5. Chicago Sun-Times, December 27, 2004 Monday, AUTOTIMES; Pg. 4, 208 words
6. Shift: Changing attitudes on vehicles and driving
7. Calgary Sun (Alberta), July 30, 2010 Friday, AUTONET; Pg. A16, 792 words, BY WHEELBASE COMMUNICATIONS
8. TRANSPORTATION UPDATE: IRS would give hybrid buyers a tax credit
9. The Oklahoman, January 19, 2006 Thursday, BUSINESS; Pg. 28, 227 words
10. Hybrid-car owners might qualify for tax credit
11. Post-Bulletin (Rochester, Minnesota), August 1, 2006 Tuesday, STATE AND REGIONAL NEWS, 337 words, Chelsey Perkins, Post-Bulletin, Rochester, Minn.
12. GREEN GLOW AT AUTO SHOW: Carmakers take aim at eco-friendly with 7 new offerings
13. Chicago Sun-Times, February 10, 2011 Thursday, NEWS; Pg. 19, 733 words, Sandra Guy, sguy@suntimes.com
14. FOLLOW ARNIE ON THE HYBRID HIGHWAY
15. The Independent (London), April 4, 2006 Tuesday, FEATURES; Pg. 3, 663 words, Edmund King
16. ON THE ROAD: Hybrids worth the money
17. San Antonio Express-News, April 11, 2008 Friday, DRIVE; Pg. 1G, 1425 words, G. Chambers Williams
18. Green machines: Reduced emissions, better mileage make hybrids popular choice
19. Farmington Daily Times (New Mexico), August 5, 2007 Sunday, BUSINESS, 868 words, — Special to The Daily Times —
20. Hybrid cars get green light from consumers
21. New Orleans CityBusiness (New Orleans, LA), June 16, 2006, NEWS, 860 words, Thomas Loggett
22. Behind the wheel
23. Monterey County Herald (California), March 3, 2006 Friday, Z-BUSINESS, 851 words, Bann M. Job
24. GM plans new kinds of hybrids in 2007
25. The Philadelphia Inquirer, November 30, 2006 Thursday, BUSINESS; Pg. D04, 463 words, From Inquirer Wire Services
26. The extra cost of buying a hybrid: \$5,000 to \$8,000 The money you'll save at the pump: About \$600 a year The opportunity to feel a little greener: Priceless; Auto makers investing in gas-electric vehicles are banking on buyers' environmental consciences. Hybrid sales have risen by 960 per cent since 2000 and that's just the beginning, JEREMY CATO writes
27. The Globe and Mail (Canada), July 26, 2005 Thursday, GLOBE MEGAWHEELS: TECHNOLOGY; Pg. G12, 1695 words, JEREMY CATO
28. Making some room for the new, guilt-free SUV: General Motors will add to its gas-electric line to include sport utilities
29. The Vancouver Sun (British Columbia), November 6, 2003 Thursday Final Edition, Business BC; Pg. D13, 511 words, John Pometto
30. The Bismarck Tribune, December 27, 2010 Monday, WIRE; Pg. 3B, 567 words, ALICIA WALLACE Daily Camera

■ **Figure 9-7** ■ Results of a search conducted in the Lexis-Nexis Academic through a library's Web site

Source: Lexis-Nexis Academic through Missouri Western State University Library.

>> Resource 5: Dictionaries, Encyclopedias, and Other General References

Sometimes you may need some general information to help you get started on a research project. In this case, you may wish to consult specialized dictionaries, handbooks, or encyclopedias. Most general encyclopedias are available in some electronic format, generally as Web-based products, such as the *Encyclopedia Britannica* online. There are, however, advantages to using a specialized subject-based encyclopedia or dictionary rather than a general one in that the articles target a more scholarly audience, assume greater subject expertise, and reference more scholarly materials in their bibliographies. Following is a list of a few specialized dictionaries, handbooks, and encyclopedias you may find in the reference collection:

Blackwell Encyclopedia of Management
CRC Handbook of Chemistry and Physics
Encyclopedia of Associations
Encyclopedia of Business Information Sources
Handbook of Industrial Engineering
Handbook of Technology and Operations
International Business Information
McGraw-Hill Encyclopedia of Science and Technology
Van Nostrand's Scientific Encyclopedia

Tanya Grant's M-Global Project

At this point, Tanya has spent many hours examining the library's online catalog, searching in online periodicals' indexes and abstracts and in newspaper indexes. She locates books and articles in scholarly and technical periodicals as well as articles in popular magazines and newspapers. Through interlibrary loans, she requests a few promising items not locally owned by the library. In the meantime, she has plenty to read and begin creating notes. She has a couple of leads to reliable Web sites from the library's online catalog and an organization she wants to research. She has a good start, and plenty of work ahead.

Searching the Web

Throughout this chapter you have seen references to the World Wide Web. The Web is the largest and fastest-growing portion of the Internet, with its appealing graphic interface, which incorporates text, images, and sound, and its ability to move from one Web page to another through hyperlinks. We next highlight some of the terms and concepts, challenges, and strategies associated with using the Web as a research tool and information source.

Fundamentals of Web Searching

Mining the Web for useful resources is always challenging and frequently frustrating, but it can yield terrific results. Why is searching such a challenge?

- The Web is huge; it contains tens of millions of documents and is growing at an astounding rate.
- The Web is constantly changing—sites appear, change, move, and disappear without warning.
- Search engines and subject directories don't work very well—they retrieve too much, they don't cover the entire Web, the relevancy ranking defies logic, and no two search engines work alike.
- The content of the Web is unregulated; anyone can add anything—fact, fiction, or fiction that looks like fact.

- There is no central index to the Web and few rules for describing Web pages.
- The process of searching, sifting through results, downloading pages, and evaluating each Web page critically is time-consuming.
- The Web is full of distractions that make it difficult to stay focused.

Despite these challenges, the Web offers access to extraordinary resources that often have no print counterpart. Because of the Web's sheer size, a search usually finds something on any topic—possibly something of value or perhaps something useless. Some studies have estimated that scholarly sites represent only 10 percent to 20 percent of the Web, but this number is still significant. Most people agree that the Web's strength lies in its information on current events, business and industry, popular culture, the government, computing, and technology, but all disciplines are represented in some way. Some resources you can find on the Web are

1. Directories of people, businesses, and organizations
2. Advertising, marketing materials, and product catalogs
3. Government documents
4. Periodicals, newspapers, and magazines
5. Books
6. Conference proceeding and reports
7. Reference tools like guides, indexes to periodicals, and dictionaries
8. An increasing number of “by subscription only” information sources
9. Sound and video clips
10. Images

>> Using Your Evaluation Skills

When you search the Web, be prepared to invest time and effort in evaluating critically what you find. Unlike books and journal articles, which undergo a rigorous editing and review process, any Web site can be loaded directly onto the Internet. You will encounter misinformation, grossly biased content, and poor text and graphic design. Evaluate Web sources using the criteria discussed earlier in this chapter. However, because Web sources do not generally follow standard publishing practices, be prepared to invest your valuable research time determining the authority, timeliness, reliability, accuracy, point of view, and validity of the source. Once you develop a systematic approach to evaluating sources, you will quickly recognize both the high- and low-quality Web sources. Be particularly alert to the following:

- **Obscured authorship:** Often a Web designer is credited as the author when in fact an organization or a corporation is the real source.
- **Out-of-date information:** The Web is littered with abandoned and unmaintained Web sites. A high-quality Web site displays the date prominently.

- **Subtle and obvious bias:** Many Web sites are elaborate advertisements promoting products, services, causes, or points of view. Data manipulation, false arguments, and unsubstantiated opinions are common.
- **Poor-quality links:** Links from a high-quality Web site usually lead you to other valuable sites; links from a poor-quality site usually lead you to other poor-quality sites. Spending time examining the links helps you determine the quality of the site.
- **Flawed style and design:** Well-organized and accessible Web sites support the research process. Although there are many cases of good research in poorly designed sites, be aware that extracting the information from overly complex sites drains away your research time.

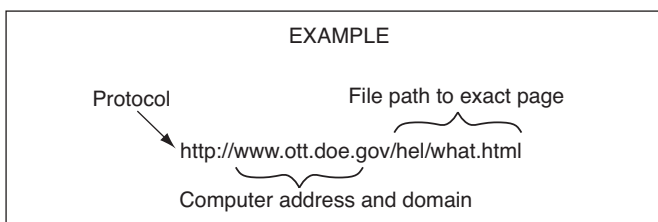
Spend time evaluating the source up front before you spend time reading the document. If you cannot determine the scope, authority, or date of the Web site, don't use it.

>> Learning the Basics

The Web is made up of millions of *Web pages*, each uniquely identified by an address or Uniform Resource Locator (URL). This *address* often contains important clues to the Web site's authorship, country of origin or domain, or the type of organization sponsoring the site. Figure 9–8 shows a list of common domains and examples.

Web browsers, such as Microsoft Internet Explorer and Mozilla Firefox, are software applications for viewing Web documents and navigating the Web. They have similar features: a line for entering URLs; options for creating *bookmark* (or *favorite*) sites; basic navigational features for moving forward and backward and stopping; and options for setting preferences to customize the browser.

com	Commercial organization (for profit)
edu	Educational institution
gov	Government organization (non-military)
int	International non-profit organizations
mil	Military organization (US)
net	Networking organization
org	Non-profit organization
ca	Canada (country of origin)
uk	United Kingdom (country of origin)



■ **Figure 9–8** ■ Common Internet domain extensions

Web Search Options

Searching the Web has become second nature for many of us. More than ever, we turn to the Web for basic news and information, to conduct business, and for entertainment. Invitations to “Visit our Web site” are everywhere, and companies have invested heavily to guarantee that their Web sites rank high in the results of a Web search. Developing effective Web research skill is critical and requires continuous updating as new techniques and search tools emerge. Options for searching include the following:

- Searching by a specific address, or URL
- Searching by keyword in an index-type *search engine* or *meta-search engine*
- Drilling through a subject category using a *subject directory*
- Using guides to reviewed and recommended Web sites

Tip: Competition among search engines is high, and new features and applications appear regularly. To keep up with search engine development and testing, try Search Engine Watch at <http://searchenginewatch.com>

Figure 9–9 lists some of the most popular search tools. Keep in mind that Web address changes or improved applications may have appeared since this list was created.

>> Searching by URL: Uniform Resource Locator

Searching by a specific Web address, or URL, is a very effective strategy, provided you have complete information and the Web page still exists. References to URLs are

■ **Figure 9–9** ■
Popular search engines and subject guides

	Web	
AltaVista	http://www.altavista.com	Keyword, directory
Ask.com	http://www.ask.com	Keyword and subject prompting using natural language
Bing	http://www.bing.com	Keyword, subject directory
Dogpile	http://www.dogpile.com	Metasearch engine
DuckDuckGo	http://www.duckduckgo.com	Keyword
Google	http://www.google.com	Keyword results based on links
Google Scholar	http://scholar.google.com	Keyword search of scholarly publications
Internet Public Library	http://www.ipl.org	Subject guide plus
Mahalo	http://www.mahalo.com	Keyword, subject All results checked by human editors
WebCrawler	http://www.webcrawler.com	Metasearch engine Keyword, subject directory
Yahoo!	http://www.yahoo.com	Keyword, subject directory

regularly included in books, journals, television and radio broadcasts, and marketing and advertising literature. One good Web site can lead you to other well-written and -maintained Web sites.

>> Searching by Keywords Using Search Engines and Meta-Search Engines

Hundreds of search engine companies on the Web have created massive databases of Web sites and provide keyword searching. Keep in mind that these search engine companies are actually in the business of selling advertising, leasing keywords, and attracting potential customers for the companies that pay to advertise.

Search engine databases are usually built without human intervention using computer programs called *robots* or *spiders* that move throughout the Web. No single search engine indexes the entire Web, and there is fierce competition among companies for the distinction of having the largest, most current, or most useful database.

The value of a search engine from a research viewpoint depends on the relevancy ranking of the results, speed, quantity, and currency of information it retrieves. The best search engines provide simple and clear instructions that allow you to refine the results. Pay particular attention to the advanced-search features.

Tip: Master the features of one search engine before moving on to the next.

Meta-search engines simultaneously use the databases of a number of search engines to respond to a request. The keyword search is forwarded to a variety of search engines; then the database results are collected and displayed. You can save time using a meta-search engine, particularly on narrow, well-defined topics, but you often lose the ability to refine a search using the features of the individual search engines.

There are hundreds of search engines, and a new and better one is always on the way. Second-generation search engines feature *intelligent agents* designed to help refine your question by providing suggestions and alternate lines of inquiry. Other second-generation search engines provide continual updating services using *push technology* that stores your search profile, runs searches, and reports results automatically.

Keeping up with developments in search engines is a challenge. Search for new ones periodically or ask colleagues to recommend one. Keep trying different ones until you find a few that meet your needs.

Tanya Grant's Web Search

Tanya begins her Web search using a URL that one of the M-Global engineers gave her, which leads her to a Web site maintained by the U.S. Environmental Protection Agency. This comprehensive site helps her organize the issues, policies, and research trends, as well as locate articles, reports, and other information sources on the subject. Next, she follows up on a reference to an organization she finds mentioned in a journal article. Using the advanced search feature in *Google*, she enters the organization's name as a phrase and locates the Website immediately. It is here that she locates a number of useful Canadian and international documents. She spends three or four hours reviewing the sites and following up the links.

Tanya spends half an hour searching for a guide to recommended and reviewed Web sites on the topic. First, she checks the library's online catalog subject guide to the World Wide Web. Although she does find a few useful guides on more general topics, there is nothing exactly on her topic. She next checks BUBL Information Service and Argus Clearinghouse for prepared guides, but neither has anything on target. Finally, she tries Ask.com and has better results. From here she is guided to the Web sites of a number of government agencies, private companies, and universities conducting studies on hybrid vehicles. She comes across a Web site for a local research center at a nearby university. She bookmarks the page and makes a note to contact the center later that day.

Tanya needs current and specific information on hybrid vehicles and tax incentives. This narrow search works well using the advanced search features of Google, Alta Vista, and Ask.com, which allow multiple domain limits such as .gov, .edu, and .org. Just to double-check, Tanya tries Dogpile, a meta-search engine; although she finds a few new sites, most of them are familiar. This is a sure sign that she has completed her Web research and should move on. Ultimately, because Tanya needs to be confident she locates the most accurate and current information, she returns to her library's online catalogs for a guide to government documents on the Web and is referred to USA.gov (www.usa.gov). Her search here is uncluttered and produces information that she can use confidently.

>>> Conducting Primary Research

Sometimes your research project may require conducting *primary research* to collect firsthand information yourself. Primary research can be defined as follows:

Primary research: Data collected by the researcher through interviews, focus groups, surveys, laboratory experiments, or field observations. Primary sources also include original works such as diaries, company reports, and correspondence, as well as documents that are the subject of analysis, such as user's manuals and Web sites.

The many ways of conducting primary research are generally divided into two methods: quantitative research and qualitative research. Surveys are often a combination of quantitative and qualitative research methods because they can present numerical data about people's opinions. Usability studies are one of the most important forms of primary research that technical communicators conduct in the workplace. This section provides an overview of basic methods of conducting primary research in technical communication.¹

¹A detailed discussion of research methods in technical communication can be found in M. A. Hughes & G. F. Hayhoe. (2008). *A research primer for technical communication: Methods, exemplars, and analysis*. New York, NY: Erlbaum. Hughes and Hayhoe's book is the source for some of the concepts in this chapter.

Quantitative Research

Quantitative research collects data that can be represented in numbers. In technical communication, this step often involves answering questions about how long it takes to perform a task, or how many clicks it takes to find information in a Help file. Technical communicators may also collect and analyze statistics from surveys and interviews. Quantitative research is judged by validity and reliability.

- Research is *valid* if it measures what it was designed to measure.
- Research is *reliable* if it can be repeated with the same results.

Qualitative Research

Qualitative research is common in technical communication. Qualitative data cannot be represented in numbers. Instead, qualitative research analyzes words, images, processes, or objects. In particular, technical communicators use the following methods to collect qualitative data:

- **Interviews.** Technical communicators often interview subject matter experts (SMEs) to learn about products or processes that they are documenting, and they should interview users to learn how to improve the usability of products or processes. Because many technical communication students conduct interviews as part of their research, this chapter provides detailed advice for preparing for interviews.
- **Focus groups.** Technical communicators may also meet with focus groups—that is, small groups of employees or clients—to learn about issues related to the design of products, Web sites, or documentation. Preparing to work with focus groups is much like preparing for interviews, although you will be recording discussion and interaction among the group members.
- **Field observations.** Technical communicators may go into the field to watch clients use equipment or software on-site, so that they can learn more about who their readers are and how their readers use equipment, software interfaces, or documentation. You should prepare for field observations by clearly identifying the goals of your research and developing a method to record and classify the information you need for your research question.
- **Document analysis.** Technical communicators may analyze documents for their quality, using theories of effective communication and usable document design. In your writing classes, you may have been asked to analyze the rhetorical or stylistic characteristics of an essay. This is one kind of document analysis.

Qualitative researchers classify and code their data to identify patterns that can help them understand the topic of their research. Qualitative research is judged by credibility, transferability, and dependability.

- Research is *credible* if the people interviewed or the processes or examples analyzed are typical of the people, processes, or examples being studied.
- Research is *transferable* if the findings can be applied to similar settings or objects.
- Research is *dependable* if different researchers would probably reach similar conclusions if they applied the same methods to similar populations, processes, or objects.



Ryan McVay/Thinkstock

Interviews

Interviews can be valuable primary sources of information in a research project. To achieve success in an interview, you must follow some common guidelines. Following are a few basic pointers for preparing, conducting, and recording the results of your interviews:

>> Step 1: Preparing for the Interview

Put at least as much effort into planning the interview as you do into conducting it. Good planning puts you at ease and shows interviewees that you value their time. Specifically, follow these guidelines:

- **Develop a list of specific objectives for the interview.** Know exactly what you want to accomplish so that you can convey this significance to the person you interview.
- **Make clear your main objectives when you make contact for the interview.** This conversation should (1) stress the uniqueness of the person's contribution, (2) put him or her at ease with your goals and the general content of the proposed discussion, and (3) set a starting time and approximate length for the interview. If handled well, this preliminary conversation will serve as a prelude to the interview, giving direction to the next meeting.
- **Prepare an interview outline.** People you interview understand your need for written reference during the interview. Indeed, they expect it of any well-prepared interviewer. A written outline should include (1) a sequential list of topics and subtopics you want to cover and (2) specific questions you plan to ask.
- **Show that you value your interviewee's time.** You can do this first by showing up a few minutes early so that the interview can begin on time. You also show this courtesy by staying on track and ending on time. Never go beyond your promised time limit unless it is absolutely clear that the person being interviewed wants to extend the conversation further than planned.

>> Step 2: Conducting the Interview

Your interview will be successful if you stay in control of it. Maintaining control has little to do with force of personality, so don't worry if you are not an especially assertive person. Instead, keep control by sticking to your outline and not letting time get away from you. If you find your interviewees straying, for example, gently bring them back to the point with another question from your list. Following are additional pointers for conducting the interview:

- **Ask mostly open-ended questions.** Open-ended questions require your respondent to say something other than yes, no, or other short answers. They are useful to the speaker because they offer an opportunity to clarify an opinion or a fact. They are useful to you because you get the chance to listen to the speaker, digest information, and prepare for the next question.

M-Global's Tanya Grant may ask questions such as "Could you describe two or three ways in which your expectations for hybrid vehicles have been met? For what purposes is your company currently using its fleet of hybrid vehicles?" or "I've been told that your company has a high commitment to environmental issues in the Atlanta area. How has purchasing and using hybrid vehicles been part of that commitment?"

- **Ask close-ended questions when you need to nail down an answer.** For example, Tanya may ask persons she interviews, "Would you be willing to meet with our fleet supervisor to discuss your experience with maintaining hybrid vehicles?" A yes or perhaps will give her an opening for calling this person several months later. A close-ended question works when commitment is needed.
- **Use summaries throughout the interview.** Brief and frequent summaries serve as important resting points during the conversation. They give you the chance to make sure you understand the answers that have been given, and they give your counterpart the chance to amplify or correct previous comments. For example, Tanya may comment to her interviewee, "So, in other words, you are saying that hybrid vehicles make most sense right now for in-city driving where only one or two people share the vehicle." This summary elicits either a yes or a clarification, either of which helps Tanya record the interview accurately.

>> Step 3: Recording the Results

You should take notes throughout the interview. The actual mechanics of this process may influence the accuracy of your note taking. Following are three possible approaches:

- **Option 1: Number reference:** Using this approach, you begin the interview with a list of numbered questions on your outline page; then, when you take notes, simply list the number of the question, followed by your notes. This approach gives you as much space as you want to write questions, but it does require that you move back and forth between your numbered question list and note page.
- **Option 2: Combined question-and-answer page:** For this approach, place a major question or two on each page, leaving the rest of the page to record answers to these and related questions that may be discussed. Although this strategy requires considerably more paper and separates your prepared list of questions, it does help you focus quickly on each specific question and answer.
- **Option 3: Split page:** Some interviewers prefer to split each page lengthwise, writing questions in the left column and corresponding answers in the right column. Some questions may have been prepared ahead of time, as in Option 2; others may be written as they are asked. In either case, you have a clear visual break between questions on one side and answers on the other. The advantage over Option 2 is that you have a visual map that shows you your progress during the conversation. Questions and answers are woven together into the fabric of your interview.

Interviews may be conducted as a follow-up to surveys. Researchers will ask to interview a few respondents to gain more detailed information about responses on surveys.

Research with Human Subjects

Much of the qualitative research that technical communicators do involves people or, in research terms, *human subjects*. If this research is being conducted through a university, you will need to comply with the institution's ethical guidelines and file the appropriate requests and reports with the Institutional Review Board (IRB). Some government research organizations and large research labs also have IRBs. In a university or other large research setting, you may be expected to file a research plan with the IRB even if your research is limited to interviews, surveys, or focus groups (although these types of research projects are usually awarded "exempt" status).

Even if your company does not have a formal IRB, it should have procedures in place to make sure that all research participants have been informed of their rights and any risks in the study, and that they have consented to participate. Participants may be asked to complete an informed-consent form like the one in Figure 9–10. Any research that may lead to publication should meet the requirements of informed consent; some journals require that article submissions be accompanied by consent forms.

Using Surveys

Surveys can combine qualitative research and quantitative research. Whereas survey results are reported in numbers, as statistics or percentages, survey questions usually ask for qualitative information, for opinions or personal experiences. They may even invite comments, which are collected, grouped, and classified. This section explains how to prepare, send out, and report the results of a survey.

Tanya Grant's M-Global Project

Recall that Tanya, who works in Marketing at M-Global, has been asked by the company president to write a report that examines the successes and failures of hybrid electric cars. This report will look at start-up costs for switching the vehicle fleet, tax and other incentives, and the potential of the technology.

Now, before reporting her findings to Jim McDuff, she wants to find out what corporate users of the technology think of its potential. She believes her best approach is to (1) send a survey to companies that have hybrid vehicle fleets and (2) personally interview three or four respondents, including employees at M-Global's Asian offices, who will help management decide on the company's direction.

Tanya Grant has the same challenge you would face in developing a survey. Like you, she receives many surveys herself. Most of them she tosses in the recycle bin because they don't warrant her time, are too long, or seem confusing. Now that the shoe is on the other foot, she wants to design a survey that attracts the attention of readers and entices them to complete it. To accomplish this feat, she goes through the following three-stage process:

>> Step 1: Preparing the Survey

Obviously, your survey is useful only if readers complete and return it. You must focus just as much on your readers' needs as you do on your own objectives. Before readers

GENERIC SAMPLE INFORMED CONSENT

Research Subject Informed Consent Form

Prospective Research Subject: Read this consent form carefully and ask as many questions as you like before you decide whether you want to participate in this research study. You are free to ask questions at any time before, during, or after your participation in this research.

This is a generic sample form to help you address most situations. Please adapt as appropriate for your research protocol and institution. *Pending rulemaking for classified human subject research will require additional elements of consent.*

Project Information	
Project Title:	Project Number:
Site IRB Number:	Sponsor:
Principal Investigator:	Organization:
Location:	Phone:
Other Investigators:	Organization:
Location	Phone:

1. PURPOSE OF THIS RESEARCH STUDY

- Include 3-5 sentences written in nontechnical language (8th grade reading level)
"You are being asked to participate in a research study designed to . . ."

2. PROCEDURES

- Describe procedures: *"You will be asked to do . . ."*
- Identify any procedures that are experimental/investigational/non-therapeutic.
- Define expected duration of subject's participation.
- Indicate type and frequency of monitoring during and after the study.

3. POSSIBLE RISKS OR DISCOMFORT

- Describe known or possible risks. If unknown, state so.
- Indicate if there are special risks to women of childbearing age; if relevant, state that study may involve risks that are currently unforeseeable, e.g., to developing fetus
- If subject's participation will continue over time, state: *"any new information developed during the study that may affect your willingness to continue participation will be communicated to you."*
- If applicable, state that a particular treatment or procedure may involve risks that are currently unforeseeable (to the subject, embryo or fetus, for example.)

4. OWNERSHIP AND DOCUMENTATION OF SPECIMENS

- Describe ownership, use, disposal, and documentation (identification) procedures for specimens or samples taken for study purposes.

■ **Figure 9-10** ■ Sample informed-consent form

Source: <http://humansubjects.energy.gov/doe-resources/files/generic-sample-informed-consent-form.doc>.

5. POSSIBLE BENEFITS

- Describe any benefits to the subject that may be reasonably expected. If the research is not of direct benefit to the participant, explain possible benefits to others.

6. FINANCIAL CONSIDERATIONS

- Explain any financial compensation involved or state: *“There is no financial compensation for your participation in this research.”*
- Describe any additional costs to the subject that might result from participation in this study.

7. AVAILABLE TREATMENT ALTERNATIVES

- If the procedure involves an experimental treatment, indicate whether other non-experimental (conventional) treatments are available and compare the relative risks (if known) of each.

8. AVAILABLE MEDICAL TREATMENT FOR ADVERSE EXPERIENCES

- *“This study involves (minimal risk) (greater than minimal risk).”* In the event that greater than minimal risk is involved, provide the subject with the following information.
- If you are injured as a direct result of taking part in this research study, emergency medical care will be provided by [name] medical staff or by transporting you to your personal doctor or medical center. Neither the [your site name] nor the Federal government will be able to provide you with long-term medical treatment or financial compensation except as may be provided through your employers insurance programs or through whatever remedies are normally available at law.

9. CONFIDENTIALITY

- Describe the extent to which confidentiality of records identifying the subject will be maintained.

“Your identity in this study will be treated as confidential. The results of the study, including laboratory or any other data, may be published for scientific purposes but will not give your name or include any identifiable references to you.”

“However, any records or data obtained as a result of your participation in this study may be inspected by the sponsor, by any relevant governmental agency (e.g., U.S. Department of Energy), by the (your site name) Institutional Review Board, or by the persons conducting this study, (provided that such inspectors are legally obligated to protect any identifiable information from public disclosure, except where disclosure is otherwise required by law or a court of competent jurisdiction. These records will be kept private in so far as permitted by law.”

In addition, list steps to protect confidentiality such as codes for identifying data.

10. TERMINATION OF RESEARCH STUDY

You are free to choose whether or not to participate in this study. There will be no penalty or loss of benefits to which you are otherwise entitled if you choose not to participate. You will be provided with any significant new findings developed during

the course of this study that may relate to or influence your willingness to continue participation. In the event you decide to discontinue your participation in the study,

- These are the potential consequences that may result: (list)
- Please notify (name, telephone no., etc.) of your decision or follow this procedure (describe), so that your participation can be orderly terminated.

In addition, your participation in the study may be terminated by the investigator without your consent under the following circumstances. (Describe) It may be necessary for the sponsor of the study to terminate the study without prior notice to, or consent of, the participants in the event that (Describe circumstances, such as loss of funding.)

11. AVAILABLE SOURCES OF INFORMATION

- Any further questions you have about this study will be answered by the Principal Investigator:

Name:

Phone Number:

- Any questions you may have about your rights as a research subject will be answered by:

Name:

Phone Number:

- In case of a research-related emergency, call:

Day Emergency Number:

Night Emergency Number:

12. AUTHORIZATION

I have read and understand this consent form, and I volunteer to participate in this research study. I understand that I will receive a copy of this form. I voluntarily choose to participate, but I understand that my consent does not take away any legal rights in the case of negligence or other legal fault of anyone who is involved in this study. I further understand that nothing in this consent form is intended to replace any applicable Federal, state, or local laws.

Participant Name (Printed or Typed):

Date:

Participant Signature:

Date:

Principal Investigator Signature:

Date:

Signature of Person Obtaining Consent:

Date:

complete a form, they must perceive that (1) it benefits them personally or professionally and (2) it is easy to fill out and return. Keep these two points in mind as you design the form and the cover letter. Following are some specific guidelines for preparing a reader-focused document:

1. **Write a precise purpose statement.** As in other documents, a one-sentence statement of purpose provides a good lead-in for your cover letter that accompanies the survey (see next section). For example, Tanya prepared the following purpose statement for her survey concerning hybrid electric vehicles: “The purpose of this survey is to find out how your experience with hybrid cars can benefit others.” As obvious as that statement sounds, it helps busy readers who don’t have time to wade through long rationales.

2. **Limit the number of questions.** Every question must serve to draw out information that relates to your purpose statement. For example, Tanya knows her questions must focus on the reader’s experience with hybrid electric vehicles. She must resist the temptation to clutter the survey with irrelevant questions on other alternative-fuel vehicles such as natural gas or electric cars.

3. **Ask mostly objective questions.** You must design your form so that (1) questions are easy to answer and (2) responses are easy to compile. Although open-ended questions yield more detailed information, the answers take time to write and are difficult to analyze. Instead, your goal is breadth, not depth, of response. With the exception of one or two open-ended questions at the end of your survey, reserve long-answer responses for personal interviews you conduct with a select audience. For example, Tanya decided to include an optional open-ended question at the end of her survey, where she asks hybrid users to recommend design improvements for hybrid electric vehicles.

Objective questions come in several forms. Four common types are described next, along with examples of each.

■ **Either/or questions:** Such questions give the reader a choice between two options, such as “yes” or “no.” They are useful only when your questions present clear, obvious choices.

Example: “Do you believe your hybrid vehicles accelerate well in all driving situations?” (followed by “yes” and “no” blocks), or “The hybrid accelerates well in all driving situations.”

■ **Multiple-choice questions:** These questions expand the range of possibilities for the reader to three or more, requiring a longer response time.

Example: “If you answered ‘yes’ to the preceding question [a question asking if the hybrid vehicle accelerates well], what is your typical driving terrain? (a) Flat; (b) Hilly; (c) Combination of flat and hilly; (d) Mountainous”

■ **Graded-Scale Questions:** By permitting degrees of response, these questions help gauge the relative strength of the reader’s opinion.

Example: “Using a hybrid vehicle has met our day-to-day driving needs. (a) Strongly agree; (b) Agree; (c) Disagree; (d) Strongly disagree; (e) Have no opinion”

■ **Short-Answer Questions:** Use these questions when the possible short answers are too numerous to list on your form.

Example: “List the makes of vehicles that your company has purchased in the last five years.”

4. **Provide clear questions that are easy to answer.** Like other forms of technical writing, surveys can frustrate readers when individual questions are unclear. Four common problems are (1) bias in phrasing, (2) use of undefined terms, (3) use of more than one variable, and (4) questions that require too much homework. Following are some examples of right and wrong ways to phrase questions, along with a brief comment on each problem:

Biased question:

Original question: “Are the federal and state government’s excessive tax credits for purchasing alternative-fueled vehicles affecting your purchasing decision?” (Words like *excessive* reflect a bias in the question, pushing a point of view and thus skewing the response.)

Revised question: “Do you believe that the federal and state tax credits affected your purchasing decision?”

Undefined technical terms:

Original question: “Are you familiar with the work of the PNGV on AFVs?” (Your reader may not know that PNGV is short for Partnership for a New Generation of Vehicles, or that AFV stands for Alternative Fuel Vehicle. Thus some “no” answers may be generated by confusion about terminology.)

Revised question: “Are you familiar with the work of the Partnership for a New Generation of Vehicles on alternative-fuel vehicles?”

Mixed variables:

Original question: “Were the dealer’s maintenance technicians prompt and thorough in their work?” (There are two questions here, one dealing with promptness and the other with thoroughness.)

Revised question: (two separate questions): “Were the dealer’s maintenance technicians prompt?” “Were the dealer’s maintenance technicians thorough?”

Question that requires too much homework:

Original question: “What other alternative fuel vehicles has your company researched, tested, or purchased in the last 10 years?” (This question asks the readers to conduct research for an accurate answer. If they do not have the time for that research, they may leave the answer blank or provide an inaccurate guess. In either case, you are not getting valid information.)

Revised question: “Has your company tried other alternative-fuel vehicles?”

5. **Include precise and concise instructions at the top of the form.** Your instructions can be in the form of an easy-to-read list of points that start with action verbs, such as the following list:

- Answer Questions 1–20 by checking the correct box.
- Answer Questions 21–30 by completing the sentences in the blanks provided.
- Return the completed form in the envelope provided by October 15, 2011.

Or if instructions are brief, they can be in the form of a short, action-centered paragraph, such as “After completing this form, please return it in the enclosed stamped envelope by October 15, 2011.”

6. **Apply principles of document design.** Although you must strive for economy of space when designing a survey, use adequate white space and other design principles to make the document attractive to the eye.

7. **Test the survey on a sample audience.** Some sort of “user test” is a must for every survey. For example, after completing her survey, Tanya decides to test it on three people:

- A fellow marketing colleague at M-Global who has conducted several surveys for the firm
- A psychologist Tanya knows through a local professional association
- A vehicle fleet manager whom she knows well enough to ask for constructive criticism on the form

Thus her user test will solicit views from people with three quite different perspectives.

>> Step 2: Conducting the Project

After you have designed a good form, the next task is to distribute it. Following are guidelines for selecting a good sampling of potential respondents, introducing the survey to your audience, and encouraging a quick response from a high percentage of readers.

1. **Choose an appropriate audience.** Selecting your audience depends on the purpose of your survey. If you manage a 100-employee engineering firm and want to gauge customer satisfaction with recent construction jobs, you might send your survey to all 156 clients you have served in the past two years. Restricting the mailing list would be unnecessary, because you have a small sample.

However, if you are in Tanya’s position at M-Global, with a mailing list totaling about 3,200 corporations that have purchased hybrid vehicles in 2010 and 2011, you must select a random sample. Tanya’s research suggests that she will receive about a 25 percent rate of return on her surveys. (Actually, this rate would be quite good for an anonymous survey.) Given that she wants about 200 returned forms, she must send out about 800 surveys in expectation of the 25 percent return rate.

With a client list of 3,200, she simply selects every fourth name from the alphabetized list to achieve a random list of 800 names. Note that the selection of client names from an alphabetized list preserves what is essential—that is, the random nature of the process.

Of course, you can create more sophisticated sampling techniques if necessary. For example, let’s assume Tanya wants an equal sampling of companies that purchased in each of the two years—2009 (with 1,200 names) and 2010 (with 2,000 names). In other words, she wants to send an equal number of forms to each year’s hybrid owners, even though the number of corporate hybrid owners varies from year to year. In this case, first she would select 400 names—or every third name—from the 1,200 alphabetized names for 2009. Then she would select the other 400 names—or every fifth name—from the 2,000 alphabetized names for 2010. As a result, she has done all she can do to equalize the return rate for two years.

This strategy helps you choose the audience for simple survey projects. You may want to consult a specialist in statistics if you face a sophisticated problem in developing an appropriate sampling.

2. **Introduce the survey with a clear and concise cover letter.** In 15 or 20 seconds, your letter of transmittal must persuade readers that the survey is worth their time. Toward this end, it should include three main sections (which correspond to the letter pattern presented in Chapter 6):

- **Opening paragraph:** State precisely the purpose of the survey and perhaps indicate why this reader was selected.
- **Middle paragraph(s):** State the importance of the project and strive to emphasize ways that it may benefit the reader.
- **Concluding paragraph:** Specify when the survey should be returned, even though this information will be included in the directions on the survey itself.

3. **Encourage a quick response.** If your survey is not anonymous, you may need to offer an incentive for respondents to submit the form by the due date. For example, you can offer to send them a report of survey results, a complimentary pamphlet or article related to their field, or even something more obviously commercial, when appropriate. Clearly, any incentive must be fitting for the context. Keep in mind also that some experts believe an incentive of any kind introduces a bias to the sample.

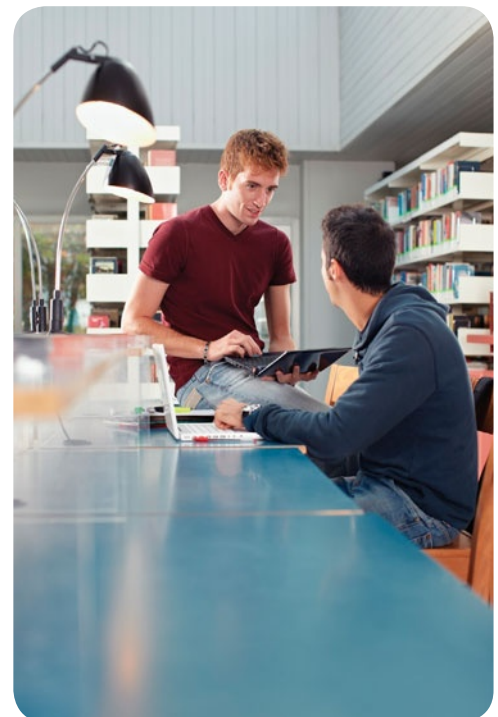
If the survey is anonymous or if complimentary gifts are inappropriate or impractical, then you must encourage a quick response simply by making the form as easy as possible to complete. Clear instructions, frequent use of white space, a limited number of questions, and other design features mentioned earlier must be your selling points.

>> Step 3: Reporting the Results

After you tabulate results of the survey, you must return to the needs of your original audience—the persons who asked you to complete the survey. They expect you to report the results of your work. Described next are the major features of such a report.

First, you must show your audience that you did a competent job of preparing, distributing, and collecting the survey; therefore the body of your report should give details about your procedures. Appendixes may include a sample form, a list of respondents, your schedule, extensive tabulated data, and other supporting information.

Second, you must reveal the results of the survey. This is where you must be especially careful. Present only those conclusions that flow clearly from data. Choose a tone that is more one of suggesting than declaring. In this way, you give readers the chance to draw their own conclusions and to feel more involved in final decision making. Graphs are an especially useful way to present statistical information (see Chapter 13).



Finally, remember that your report and the completed surveys may remain on file for later reference by employees who know nothing about your project. Be sure that your document is self-contained. Later readers who uncover the “time capsule” of your project should be able to understand its procedures and significance from the report you have written.

Usability Testing

Usability typically involves setting goals, selecting criteria, developing test materials, soliciting participants, setting up the testing environment, conducting the test, and writing a results report. For more information on formal usability testing, three useful books are Jakob Nielsen’s *Usability Engineering*, Jeffery Rubin’s *Handbook of Usability: How To Plan, Design, and Conduct Effective Tests*, and Carol Barnum’s *Usability Testing Essentials: Ready, Set, Test*. Usability testing is such an important form of research that some companies employ full-time usability testers and have well-equipped usability labs. Like surveys, usability testing can combine quantitative research and qualitative research.

Almost any product or process can be tested for usability, but this chapter will focus on usability testing of print and digital documents. Usability testing of Web sites is discussed in Chapter 14. Procedural documents, including user guides, instructions, and Help files should all be tested for how usable they are. Procedures are considered usable if they are

- Easy to learn
- Efficient to use
- Easy to remember

The first step in usability testing, as in other research, is to identify the goal of the research. Usability testing can answer questions such as the following:

- How clear are the instructions?
- How quickly can a user find information in a Help file?
- How useful are the illustrations in a user’s guide?

Although many characteristics of a document may be tested, it is best to test only one characteristic at a time.

A usability-testing lab allows testers to observe and measure how actual users interact with objects, software, Web sites, or documents. These interactions may be monitored with cameras or one-way mirrors, or computers may record key strokes to see how users try to access information. Some labs even have equipment that allows testers to record users’ eye movements as they look for information on a computer screen.

While quantifiable characteristics, such as the number of clicks to complete a task, can be measured, it is just as important to measure users’ satisfaction with a product or document, that is, to measure how usable they *perceive* the document to be. Surveys, interviews, and focus groups can help technical communicators learn how users feel about their interactions with a product or document. One low-tech way of testing user interaction with a document is a *think-aloud protocol*. In this method, the user works through

a process such as finding information in a Help file or learning a new software program while speaking his thoughts aloud. The tester records the thoughts and makes notes about the user's actions. As in other qualitative research, it is important to sort the data gathered by these methods, and to classify them in ways that reveal how usable the item is.

>>> Using Borrowed Information Correctly

In some workplace writing, issues of citation can become complicated, especially in collaborative projects that use documents published by the writer's organization, and that will be published under the organization's name. (See Chapter 3 for more on collaborative writing.) However, whenever you are using material that has been published in a book, periodical, or on another organization's Website, you should cite your sources.

Most errors in research papers occur in transferring borrowed information. This section has three goals: (1) to explain why you must acknowledge sources you have used, (2) to outline a research process from the point at which you identify sources of information, and (3) to provide sample documentation styles from three well-known style manuals.

Avoiding Plagiarism

One basic rule underlies the mechanical steps described in the rest of this chapter:

With the exception of common knowledge, you should cite sources for all borrowed information used in your final document, including quotations, paraphrases, and summaries.

Common knowledge is information generally available from basic sources in the field. In the case of Tanya's research project, *common knowledge* is a definition of hybrid electric vehicles. When you are uncertain whether a piece of borrowed information is common knowledge, go ahead and cite the source. It is better to err on the side of excessive documentation than to leave out a citation and risk a charge of *plagiarism* (the intentional or unintentional use of the ideas of others as your own). Following are three main reasons for documenting sources thoroughly and accurately:

1. **Courtesy:** You owe readers the *courtesy* of citing sources where they can seek additional information on the subject. Sources should be given for quotations, paraphrases, and summaries.
2. **Ethics:** You have an *ethical* obligation to show your reader where your ideas stop and those of another person begin; otherwise, you are parading the ideas of others as your own.
3. **Law:** You have a *legal* obligation to acknowledge information borrowed from a copyrighted source. In fact, you should seek written permission for the use of borrowed information that is copyrighted when you plan to publish your document or when you are using your document to bring in profit to your firm (as in a proposal or report). If you need more specific information about copyright laws or about the legalities of documentation, see a research librarian.

Certainly some plagiarism occurs when unscrupulous writers intentionally copy the writing of others without acknowledging sources. However, most plagiarism results from sloppy work during the research and writing process. Described next are two common types of unintentional plagiarism. Although the errors are unintentional—that is, the writer did not intend to cheat—both result in the unacknowledged use of another person's work. That's plagiarism.

Mike Pierson, a supervisor at M-Global's Cleveland office, has been asked to deliver a presentation at an upcoming conference on hybrid electric vehicles. In his last-minute rush to complete the presentation—which will be published in a collection of papers from the meeting—Mike is taking notes from a source in the company library. He hurriedly writes notes from a source on a note card but fails to indicate the source. Later, when he is writing the paper draft, he finds the card and does not know whether it contains information that was borrowed from a source or ideas that came to him during the research process. If he incorporates the passage into his paper without a source, he will have committed plagiarism.

In our second case, Mike transfers a direct quotation from a source into a computer document file but forgets to include quotation marks. If he were to incorporate the quotation into his presentation later *with* the source citation but *without* quotation marks, he would have plagiarized. Why? Because he would be presenting the exact words of another writer as his own paraphrase. The passage would give the appearance of being his own words that are supported by the ideas of another, when in fact the passage is a direct quote. Again, remember that the test for plagiarism is not one's intent; it is the result.

The next section shows you how to avoid plagiarism by completing the research process carefully. In particular, it focuses on a methodical process that involves (1) bibliography notes, (2) a rough outline, (3) notes of three main kinds, (4) a final outline, and (5) drafts.

Selecting and Following a Documentation System

Documentation refers to the mechanical system you use to cite sources from which you borrow information. This section briefly compares documentation styles from three important style manuals—from the previously mentioned APA, the Modern Language Association (MLA), and the Council of Science Editors (CSE)—and provides examples for the most common citations. For complete details about a particular documentation system you are using, consult one of the manuals in the list that follows or consult the Web site of the organization that publishes the manual. Pay special attention to new guidelines these manuals may provide for documenting information from online databases and the Internet.

There are almost as many styles for documenting research as there are professional organizations, but all have the same goal of showing readers the sources from which you gathered information. One of your early steps in research is to determine which style manual to use. Often your instructors select a discipline-specific style manual. Style manuals guide the writer through the editorial rules governing everything from use of headers and pagination and graphic and text layout to managing data display and, of course, the rules for documenting sources. Style manuals are regularly revised by the organizations that publish them. One of the areas of greatest changes is the rules for citing electronic resources. As the variety and use of electronic materials continue to evolve, so, too, do the style manuals. Be sure to check the edition of the style manual you are using to make sure it is the latest available.

Following are just a few documentation manuals commonly used in business, industry, and the professions. You can often locate useful tips and examples at the Web sites maintained by each of these organizations in addition to the purchasing information or the style manual itself.

American Psychological Association (APA)

Publication Manual of the American Psychological Association, 6th ed. 2010.

Council of Science Editors (CSE)

Scientific Style and Format: The CSE Manual for Authors, Editors, and Publishers, 7th ed. 2006.

Modern Language Association (MLA)

MLA Handbook for Writers of Research Papers, 7th ed. 2009.

University of Chicago Press

A Manual for Writers of Research Papers, Theses, and Dissertations, 7th ed., 2007.

University of Chicago Press

Chicago Manual of Style, 16th ed. 2010. Also noteworthy:

University of Wisconsin's Writing Center

Writer's Handbook

Web site: <http://www.wisc.edu/writing/Handbook>

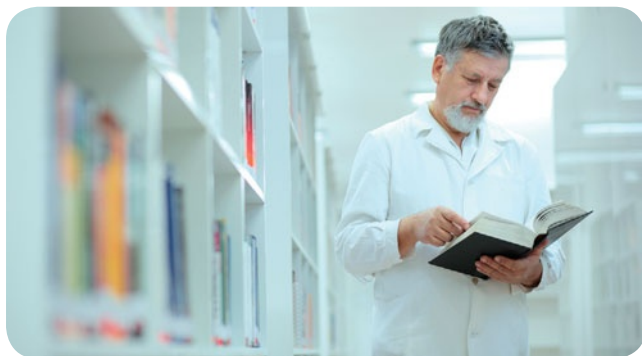
Purdue Online Writing Lab

Web site: <http://owl.english.purdue.edu>

We focus briefly on the APA, MLA, and CSE manuals and compare documentation styles for citing works. The three systems share some characteristics. Each uses parenthetical references in the body of the report that lead the reader to a separate works-cited or reference page. Each system cites the author's name and either the publication year (APA and CSE) or the relevant page number where the fact, quote, or observation can be located (MLA). Frequently, the content of the parenthetical references is blended into the text with perhaps only the date or page in parentheses. The works-cited or reference page is arranged alphabetically by the author's last name for APA and MLA. CSE uses a numbered bibliography system.

CSE offers three style choices: the *name-year* system, the *citation-sequence* system, and the *citation-name* system. The name-year system is similar to APA style, using a parenthetical reference to the date. In the citation-sequence system and the citation-name system, the parenthetical citation refers to a numbered list of citations at the end of the document. For the citation-sequence system, the sources in the bibliography are numbered sequentially in the order in which they appear in the document. In the citation-name system, the sources in the bibliography are alphabetized by the authors' last names, and then numbered in that order. When using CSE, you must determine which system is preferred—the name-year system, the citation-sequence system, or the citation-author system. Check with your instructor or editor.

There are significant and subtle variations in parenthetical entries and workscited listings when the style manuals are closely compared. The Handbook in Appendix A offers a few basic examples. Writers must consult the style manual itself for a thorough discussion.



lightpoet/Shutterstock

>>> Reporting Your Research

Although you may occasionally conduct workplace research for your own use, you will usually be expected to share your results with others. The written formats for sharing research include reports, which are discussed in Chapters 10 and 11; proposals and white papers, which are discussed in Chapter 12; and presentations within your organization, to clients, or to other members of your profession, which are discussed in Chapter 15.

ABC Format for Technical Research

The ABC format offers an effective way to organize your presentation of your research, whether you are presenting it in a report, a proposal, an article in a professional magazine or journal, or even a presentation at a professional conference. The abstract identifies the problem you are discussing and provides your reader the background for the problem that

you are discussing. This should include a review of the published research, often referred to as a *literature review*. Many of the research papers you write for school will be literature reviews of articles and books you have read about a topic. The body explains your methodology, or how you gathered data from your primary sources, and presents, analyzes, and discusses your results. The conclusion identifies your most important findings. It may also recommend actions to be taken, or it may recommend further research.

See Model 9–1 on pages 295–299 for Tanya Grant’s complete memo report, which cites research.

ABC Format: Research

- **ABSTRACT:** Provides the background the reader needs to understand and evaluate the research.
 - Identifies the question, problem, or issue being researched.
 - Reviews the published research about the topic, or secondary sources.
 - Overviews the organizational plan of the document.
- **BODY:** Presents and discusses the findings.
 - Explains the methodology used to gather information from primary sources.
 - Presents the results, using tables, charts, and graphs as necessary.
 - Interprets the findings through analysis and discussion.
- **CONCLUSION:** Identifies the most important findings and explains the implications of the findings.
 - May include recommendations.
 - May make a prediction.

Writing Research Abstracts

The term *abstract* has been used throughout this book to describe the summary component of any technical document. As the first part of the ABC pattern, it gives decision makers the most important information they need. However, here we use *abstract* for a narrower purpose: It is a stand-alone summary that provides readers with a capsule version of a piece of research, such as an article or a book. This section (1) describes the two main types of research abstracts, with examples of each and (2) gives five guidelines for writing research abstracts.

Types of Abstracts

There are two types of abstracts: informational and descriptive. As the following definitions indicate, informational abstracts include more detail than descriptive abstracts:

Informational Abstract

- **Format:** This type of abstract includes the major points from the original document.
- **Purpose:** Given their level of detail, informational abstracts give readers enough information to grasp the main findings, conclusions, and recommendations of the original document.
- **Length:** Although longer than descriptive abstracts, informational abstracts are still best kept to one to three paragraphs.
- **Example:** A sentence from such an abstract might read, “The article notes that functional résumés should include a career objective, academic experience, and a list of the applicant’s skills.” (See corresponding example in definition of a descriptive abstract.)

Descriptive Abstract

- **Format:** This type of abstract gives only main topics of the document, without supplying supporting details such as findings, conclusions, or recommendations.
- **Purpose:** Given their lack of detail, descriptive abstracts can help readers decide only whether they want to read the original document.
- **Length:** Their lack of detail usually ensures that descriptive abstracts are no more than one paragraph.
- **Example:** A sentence from such an abstract might read, “The article lists the main parts of the functional résumé.” (See corresponding example in definition of an informational abstract.)

You may wonder when you’ll need to write abstracts during your career. First, your boss may ask you to summarize some research, perhaps because he or she lacks your technical background. Second, you may want to collect abstracts as part of your own research project. In either case, you must write abstracts that reflect the tone and content of the original document accurately.

Assume, for example, that your M-Global supervisor asked you to read some influential research on information design. Later, your boss plans to use your abstracts to get an overview of the field and to decide which, if any, of the original full-length documents should be read in full. The examples that follow show both informational and descriptive abstracts of an article by Janice Redish. The informational abstract appeared in a bibliographic article that listed important publications about technical communication. The descriptive abstract appeared at the beginning of Redish’s article in the journal *Technical Communication*. Note that the informational abstract summarizes Redish’s findings, whereas the descriptive abstract lists two key points in the article.

Informational Abstract:

Redish, Janice C. 2000. “What is information design?” *Technical communication* 47, no. 2: 163–166.

Redish offers two meanings of information design: “the overall process of developing a successful document” and “the way the information is presented on the page or screen” (p. 163). In either case, Redish observes, the objective is “to develop a document (or communication) that works for its users” through considering the users’ needs, their ability to understand what they find, and their capacity to use their findings (p. 163). The author indicates four vital concerns in information design: planning questions and front-end analysis; iterative evaluation; the interaction and equal importance of writing and presentation; and planning question-based guidelines for design purposes (p. 163). The two critical trends in technical communication Redish indicates are the Web and single sourcing; the visual aspects of the former and the multiple uses of the latter constrain information design, and communicators must consider the “whole”—process and product, writing and design—to create successful documents.²

Descriptive Abstract:

1. Defines two meanings of information design: the overall process and the presentation of information on page and on screen
2. Predicts the future importance of both meanings of information design, in terms of design for the Web and single-sourcing³

Guidelines for Writing Research Abstracts

The following guidelines help you (1) locate the important information in a document written by you or someone else and (2) present it with clarity and precision in an abstract. In every case, you must present a capsule version of the document in language the reader can understand. The ultimate goal is to save the readers’ time.

>> Abstracting Guideline 1: Highlight the Main Points

This guideline applies whether you are abstracting a document written by you or one written by someone else. To extract information to be used in your abstract, follow these steps:

1. Find a purpose statement in the first few paragraphs.
2. Skim the entire piece quickly, getting a sense of its organization.
3. Read the piece more carefully, underlining main points and placing comments in margins.
4. Pay special attention to information gained from headings, first sentences of paragraphs, listings, graphics, and beginning and ending sections.

>> Abstracting Guideline 2: Sketch an Outline

From the notes and marginal comments gathered in Abstracting Guideline 1, write a brief outline that contains the main points of the piece. If you are dealing with a well-organized piece of writing, it is an easy task; if not, it is a challenge.

²G. J. Alred. (2003). Essential works on technical communication. *Technical Communication*, 50(4), 585–616.

³J. C. Redish. (2000). What is information design? *Technical Communication*, 47(2), 163–166.

>> Abstracting Guideline 3: Begin with a Short Purpose Statement

Both descriptive and informational abstracts should start with a concise overview sentence. This sentence acquaints the reader with the document's main purpose. Stylistically, it should include an action verb and a clear subject. Following are three options that can be adapted to any abstract:

- The article "Recycle Now!" states that Georgia must intensify its effort to recycle all types of waste.
- In "Recycle Now!" Laurie Hellman claims that Georgia must intensify its effort to recycle all types of waste.
- According to "Recycle Now!" Georgians must intensify their efforts to recycle all types of waste.

>> Abstracting Guideline 4: Maintain a Fluid Style

One potential hazard of the abstracting process is that you may produce disjointed and awkward paragraphs. You can reduce the possibility of this stylistic flaw by following these steps:

- Write in complete sentences, without deleting articles (*a, an, the*)
- Use transitional words and phrases between sentences
- Follow the natural logic and flow of the original document itself

>> Abstracting Guideline 5: Avoid Technical Terms Readers May Not Know

Another potential hazard is that the abstract writer, in pursuit of brevity, will use terms unfamiliar to the readers of the abstract. This flaw is especially bothersome to readers who do not have access to the original document. As a general rule, use no technical terms that may be unclear to your intended audience. If a term or two are needed, provide a brief definition in the abstract itself.

Note, also, that abstracts that might become separated from the original document should include a bibliographic citation.

Abstract Guidelines

- Highlight the main points
- Sketch an outline
- Begin with a short purpose statement
- Maintain a fluid style
- Avoid technical terms readers may not know

>>> Chapter Summary

- Research projects in the workplace aim to answer questions, make decisions, or solve problems.
- All research projects start with a question to be answered. The goals of the research should be clear before any sources are consulted.
- Secondary sources include research results that have been published or shared with the public in some format.

- All research projects should start with a review of the published (secondary) research.
- You can locate published research, or secondary sources, by searching library databases and the World Wide Web.
- Primary sources result in data that the researcher has gathered firsthand.
- Quantitative research collects data that can be represented in numbers.
- Quantitative research is judged by its validity and reliability.
- Qualitative research presents nonnumerical data in words or images.
- Qualitative research is judged by its credibility, transferability, and dependability.
- Collecting data from people, or, in research terms, from human subjects, requires the researcher to follow established ethical guidelines and receive the informed consent of research participants.
- Surveys must be carefully designed and tested to ensure that their results apply clearly to the research project.
- Usability testing, an important type of research for technical communicators, helps determine if products, processes, and documents are easy to learn, use, and remember. They also measure user satisfaction.
- Ethically and legally, it is important for researchers to use information borrowed from sources correctly.
- Using the correct documentation system consistently helps readers understand the information in a research paper, and it identifies the researcher as a member of a professional community.
- The ABC format helps researchers report their findings in a clear, well-organized way.
- Research abstracts provide readers with useful summaries and help readers decide if they want to read an article.

>>> Learning Portfolio

Communication Challenge To Cite or Not to Cite

Dan Gibbs works as a benefits and finance specialist at M-Global's corporate office in Baltimore. As the number of M-Global employees has grown, he has received many inquiries about ways to save for retirement. He recently wrote and distributed a four-page flyer on the topic using materials from print and online sources. The response was so positive that his boss wants to send the flyer to clients as a "freebie"—both to help clients' employees and to create good will in marketing. This use of the flyer has made Dan rethink how he developed the piece. This case study presents Dan's research process and his results. It ends with questions and comments for discussion and an assignment for a written response to the Challenge.

Background of Retirement Booklet

Unlike Tanya Grant in the hybrid vehicle project described in this chapter, Dan didn't have time or interest in pursuing a full-scale library search about retirement strategies. Besides, he has personnel magazines in the office with data that support his points. In addition, he has access to databases of relevant information through the Internet.

Like many companies, M-Global has a retirement plan largely in the form of what is called a 401k program. It allows employees to contribute a percentage of their salaries into a tax-deferred retirement account, a portion of which is matched by the employer. Even though M-Global has a generous matching arrangement, many employees do not take full advantage of the program. Therefore, Dan wrote the retirement flyer to remind them that it is never too early to plan for retirement. As it happens, he learned that many U.S. workers are failing to put away enough money for their retirement years.

The Research Process

After outlining his goals for the booklet, Dan began surfing through related information on the Internet. He made use of three sources he found on the Internet. Following are three of the themes he stressed, along with related information he used from an article in a personnel magazine.⁴

1. **Theme 1:** *We're living longer past retirement.* The following changes occurred in years of life expected after age 65: for men, 12.8 in 1960, 13.1 in 1970, 14.1 in 1980, 15.1 in 1990, and 16.0 in 2000; for women, 15.8 in 1960, 17.0 in 1970, 18.3 in 1980, 18.9 in 1990, and 19.0 in 2000.
2. **Theme 2:** *We cannot depend exclusively on Social Security.* As many more people retire from the baby boom generation born between 1946 and 1964, fewer workers paying Social Security are supporting each person getting it. The following numbers are actual and projected number of workers supporting each retiree: 7.11 in 1950, 5.67 in 1960, 5.36 in 1970, 5.04 in 1980, 4.70 in 1990, 4.65 in 2000, 4.49 in 2010, 3.45 in 2020, 2.67 in 2030, and 2.61 in 2040.
3. **Theme 3:** *We should begin saving when we're young.* If you start saving \$100 a month in a tax-deferred account at age 22, with an 8 percent annual return, you'll accumulate \$450,478 by age 65. If you start at age 32, you'll have \$194,654 by age 65.

Questions and Comments for Discussion

1. If you were presenting the previously mentioned data in a research report, what format would you choose? Why? (See Chapter 13.)
2. Considering the data sets Dan took from the magazine article/Internet sources, which ones need documentation and which, if any, do not? Explain your answer.
3. Does the fact that the flyer will be sent to clients have any effect on your answer to Question 2?
4. Do an APA-style works-cited reference for the data in Theme 1 and Theme 3. See the footnote to this Communication Challenge for actual source information. You may need to access the Web site and find the magazine article in your library's databases. Be prepared to discuss what challenges you had in formatting the citations.

Write About It

Using your library periodical databases, find an article that explains 401k retirement plans. Write an informational abstract of the article; and include a full citation of the article in APA style.

⁴Source for item 1: U.S. Department of Health and Human Services. (2010). Health, United States, <http://www.cdc.gov/nchs/data/health/hus10.pdf>. Source for item 2: Workers per Retiree: 1950–2050. <http://www.econdataus.com/workers.html>. Source for item 3: M. B. Franklin. (2008, February 1). 6 Simple Ways to Retire Rich. *Kiplinger's Personal Finance*, 54–62.

Collaboration at Work Surfing the Turf

General Instructions

Each Collaboration at Work exercise applies strategies for working in teams to chapter topics. The exercise assumes you (1) have been divided into teams of about three to six students, (2) will use time inside or outside of class to complete the case, and (3) will produce an oral or written response. For guidelines about writing in teams, refer to Chapter 3.

Background for Assignment

The Internet has greatly expanded the range of information you can find on a subject and the speed with which it can be found. Yet the Internet also has introduced new challenges. While conducting this type of research, you must do the following:

- Stay focused so that the inevitable distractions of new links and fascinating data do not detract from your main purpose.
- Evaluate the reliability of sources that are quite different from traditional hard-copy sources found in a library.
- Determine the mix of Internet and library sources that provide the best support for your topic.

- Keep good notes so that later you can properly document information that has been secured from the Internet.

To start you thinking about the process of Internet research, this exercise asks you to work with your team on a short project.

Team Assignment

First, agree as a team on a topic from the list below that you want to find information for on the Internet.

- Content management systems
- Document design
- Information design
- Technical illustrations
- Technical writing
- Usability testing

Second, work individually to locate three to five sources of information on the topic (the information itself—not just a list of sources). Third, come back together as a team and discuss the relative value of the sources of the information you found.

Assignments

Assignments can be completed either as individual exercises or as team projects, depending on the directions of your instructor. Your instructor will ask you to prepare a response that can be delivered as an oral presentation for discussion in class. Analyze the context of each Assignment by considering what you learned in Chapter 1 about the context of technical writing, and answer the following questions:

- What is the purpose of the document to be written?
- What result do you hope to achieve by writing it?
- Who are your readers and what do they want from your document?
- What method of organization is most useful?

If your instructor considers it appropriate, use a copy of the Planning Form at the end of the book for completing these assignments.

1. Analysis: Journal Article

Find a journal in your major field. Identify the following sections in one of the journal's articles:

- Review of literature
- Methodology
- Results
- Analysis and discussion
- Conclusion

Do the articles in the journal include headings? Abstracts? Illustrations? Look for information for authors about submitting articles. (This information may be on the journal's Web site.) Does the journal have specific formatting requirements for headings, captions, or other elements of the articles? What documentation style does the journal request?

2. Analysis: Research Methods

Find a journal in your major field. Your instructor may ask you to use the same article that you analyzed in Assignment 1. Locate the methodology section of the article. Does the article use quantitative methods, qualitative methods, or a combination of the two? How does the author assure the reader that the data reported in the article meet the accepted standards for the methodology? (See pages 271 and 274.)

3. Analysis: Technical Communication Database

One of the most useful databases of sources in technical communication is <http://tc.eserver.org>. One of the features of this Web site is the *tag cloud*. The link to the tag cloud is located on the lower part of the middle of the page. Open the tag cloud and study it. Look around the tc.eserver Web site to find out other ways of searching for information. Write a paragraph explaining how to find information on tc.eserver, or be prepared to discuss the site in class.

4. Analysis: Survey

Using the guidelines in this chapter for surveys, point out problems posed by the following questions:

- A. Is the poor economy affecting your opinion about the current Congress?
- B. Do you think the company's severe morale problem is being caused by excessive layoffs?
- C. Was the response of our salespeople both courteous and efficient?
- D. Of all the computer consultants you have used in the previous 15 years, which category most accurately reflects your ranking of our firm: (a) the top 5 percent, (b) the top 10 percent, (c) the top 25 percent, (d) the top 50 percent, or (e) the bottom 50 percent?
- E. In choosing your next writing consultant, would you consider seeking the advice of a professional association such as the STC or the CPTSC?
- F. Besides the position just filled, how many job openings at your firm have been handled by Dowry Personnel Services?

5. Practice: Research Paper

Using a topic approved by your instructor, follow the procedure suggested in this chapter for writing a paper that results from some technical research. Be sure that your topic (1) relates to a technical field in which you have an interest, by virtue of your career or academic experience; and (2) is in a field about which you can find information in nearby libraries.

6. Practice, M-Global Context: Research Paper

As an M-Global engineer or scientist, you have been asked to write a research paper for M-Global's upper management. Choose your topic from one of the technical fields listed below. Assume that your readers are gathering information about the topic because they may want to conduct consulting work for companies or government agencies involved in these fields. Focus on advantages and disadvantages associated with the particular technology you choose. Follow the procedure outlined in this chapter.

- Artificial intelligence
- Chemical hazards in the home
- Fiber optics
- Forestry management
- Geothermal energy
- Human-powered vehicles
- Lignite-coal mining
- Organic farming
- Satellite surveying
- Solar power
- Wind power

7. Practice: Abstract—One Article or Several Articles

- Option A** Visit your college library and find a magazine or journal in a technical area, perhaps your major field. Then photocopy a short article (about five pages) that does not already contain a separate abstract or summary at the beginning of the article. Using the guidelines in this chapter, write both an informative and a descriptive abstract for a nontechnical audience. Submit the two abstracts, along with the copy of the article.
- Option B** Follow the instructions in option A, but use a short article that has been selected or provided by your instructor.
- Option C** Read three to five current articles in your major field. Write an abstract that summarizes all of them on one page.

8. Practice: Writing a Survey

Design a brief survey to be completed by students on your campus. Select a topic of general interest, such as the special needs of evening students or the level of satisfaction with certain college facilities or services. Administer the survey to at least 20 individuals (in classes, at the student union, in dormitories, etc.). After you analyze the results, write a brief report that summarizes your findings. *Note:* Before completing this exercise, make sure that you gain any necessary approval by college officials, if required.

9. Practice: Interview

Select a simple research project that would benefit from information gained from an interview. (Your project may or may not be associated with a written assignment in this course.) Using the suggestions in this chapter, conduct the interview with the appropriate person.

10. Practice: Usability Test

Choose a simple, specific task for using a computer program that you have access to, for example, changing paragraph format. Identify the aspect of usability that you will test, such as how long it takes a user to complete the task, how many errors a user makes while trying to complete the task, or how many clicks it takes a user to find information in a Help file. Practice the task several times yourself to determine the criteria for a successful interface. How many minutes? How few errors? How many clicks?

Pair up with a class member and administer your usability test, recording your data. Your instructor may ask you to include a think-aloud protocol in your test. Write a brief report of your results, including whether the interface was successful for your user.

11. Ethics Assignment

This assignment is best completed as a team exercise.

Assume your team has been chosen to develop a Web-based course in technical communication. Team members are assembling materials on a Web site that can be used by students like you—materials such as (1) guidelines and examples from this book, (2) scholarly articles on communication, (3) newspaper articles and graphics from print and online sources, and (4) examples of technical writing that have been borrowed from various engineering firms.

Your team has been told that generally speaking, the “fair use” provision of the Copyright Act permits use of limited amounts of photocopied material from copyrighted sources without the need to seek permission from, or provide payment to, the authors—as long as use is related to a not-for-profit organization, such as a college. Your tasks are as follows:

A. Research the Copyright Act to make sure you understand its application to conventional classroom use. If possible, also locate any guidelines that relate to the Internet.

- B.** Develop a list of some specific borrowed materials your team wants to include on the site for the technical communication course. These materials may fall inside or outside the four general groupings noted previously.
- C.** Discuss how the medium of the Internet may influence the degree to which the fair use provision is applicable to your Web course. Be specific about the various potential uses of the material.
- D.** Consult an actual Web-based college course in any field and evaluate the degree to which you think it follows legal and ethical guidelines for usage.
- E.** Prepare a report on your findings (written or oral, depending on the directions you have been given by your instructor).



12. International Communication Assignment

Using interviews, books, periodicals, or the Internet, investigate the degree to which writers in one or more cultures besides your own acknowledge borrowed information in research documents. For example, you may want to seek answers to one or more of the following questions: Do you believe acknowledging the assistance of others is a matter of absolute ethics, or should such issues be considered relative and therefore influenced by the culture in which they arise? For example, would a culture that highly values teamwork and group consensus take a more lenient attitude toward acknowledging the work of others? These are not simple questions. Think them through carefully.



13. A.C.T. N.O.W. Assignment (Applying Communication To Nurture Our World)

Interview two or three students to find out why they do or do not participate in student elections on campus. On the basis of information you gather from the interviews, develop a survey form by which you systematically solicit information on the topic from a wider audience. Administer the survey to at least 10 students and include the results in an oral or written report, depending on the instructions you are given.

MEMORANDUM

DATE: May 7, 2011
TO: Jim McDuff, President
FROM: Tanya Grant, Technical Writer
SUBJECT: Hybrid Vehicle Research and Recommendations

INTRODUCTION

When you heard that our Tokyo office had purchased some Toyota vehicles powered by a combination gasoline-electric engine, you asked me to research the hybrid vehicles and to recommend whether or not M-Global should use them in our offices in the United States. After making several phone calls, checking useful Web sites, and reviewing several magazine and newspaper articles, I recommend that M-Global replace our conventional company vehicles with the more fuel-economic and energy-efficient hybrid vehicles.

ADVANTAGES

Hybrid vehicles offer several advantages, especially in cost and impact on the environment.

• Excellent Fuel Economy

Hybrid vehicles are able to double the fuel economy of many of today's conventional cars. The midsize Toyota Prius is rated by the U.S. Environmental Protection Agency at 51 miles per gallon (mpg) city driving, and at 48 mpg on the highway. According to the Environmental Protection Agency, because of regenerative braking, electric motor drive/assist, and automatic start/shutoff, hybrid vehicles are able to use stop-and-go traffic to their advantage by conserving and regenerating energy.

• Low Emissions

In a hybrid vehicle, the electric motor releases no emissions, and the small gasoline engine recharges the battery and has to work less often, so its emissions are much lower. On average, hybrid cars produce about 90 percent fewer harmful pollutants and greenhouse gases than comparable gasoline cars, according to Hybrid-Car.org (n.d.).

As hybrid technology advances over the next few years, additional hybrid vehicles, such as heavy-duty trucks and plug-in hybrids, could also be added to the fleet in order to help reduce the number of greenhouse gases released into our atmosphere. A recent study shows that conventional vehicles average 452 grams of CO₂ emissions per mile, while plug-in hybrid vehicles average only 294 grams of CO₂ emissions per mile (Richard, 2008). And according to a study performed by the Northeast Advanced Vehicle Consortium (NAVC; 2010), heavy-duty hybrid vehicles produce about 50 to 70 percent less particulate matter, and therefore a smaller amount of hazardous airborne particles, like carbon.

• Long Range

Because the gasoline engine automatically recharges the electrical motor, hybrid vehicles can go hundreds of miles before refueling due to their excellent fuel economy. With a fuel tank of 11.9 gallons, the 2011 Toyota Prius can easily go over 500 miles between gas stations (Toyota, 2011).

Research question:
Should we use hybrid
vehicles in our offices in
the United States?
Methods used to gather
information

- **Conventional Fueling**

Hybrid vehicles run on gasoline, so fuel is no different than for a standard automobile.

- **Adequate Power**

Some power is lost between the conventional and hybrid versions, yet both have adequate power. The conventional 2011 Chevy Malibu has a four-cylinder engine that holds 169 horsepower at 6400 rpm, and the hybrid Malibu holds 164 horsepower. Both can exceed 100 mph, but the hybrid Malibu averages about seven more miles per gallon (Chevrolet, 2011a).

DISADVANTAGES

Hybrid technology is relatively new, so there are still questions and concerns about it. Two disadvantages are especially important.

- **Electromagnetic Field (EMF) Risk Still Uncertain**

Although EMFs are all around us in cell phones, microwaves, televisions, and utility lines, the risk of exposure from hybrids is still uncertain. The kind of EMF most likely to be found in a hybrid is considered to be of a lower frequency—which means that it dissipates at a very short distance from the source (HybridCars.com, 2006). In a company statement, Toyota said. “The measured electromagnetic fields inside and outside of Toyota hybrid vehicles in the 50 to 60 hertz range are at the same low levels as conventional gasoline vehicles. Therefore there are no additional health risks to drivers, passengers or bystanders” (Motivalli, 2008).

- **Complexity**

With two power trains, hybrids are far more complex than conventional cars, and that could mean more down time and higher maintenance costs. However, with the recent rise in production of hybrid vehicles, there has also been a rise in the number of certified technicians that can perform the maintenance on them.

- **Battery**

Since hybrid vehicles are electric, the life of the battery becomes a concern. In the past hybrid batteries have taken up to 14 hours to charge. However, the Nissan Leaf battery is now able to be charged in 4 to 8 hours using a 220-volt home unit, or it will charge up to 80 percent in 26 minutes using a Nissan “quick charge station.” The only issue is that these stations do not exist yet. One option for M-Global, once the “quick charge stations” are operational, would be to build stations for the company’s personal use. An electrician can install a station for under \$35,000 (Dworetzky, 2010).

TAX AND LEGISLATIVE ADVANTAGES

- **Tax Credits**

State

Right now, about 35 states offer tax incentives (up to \$3,000) for the purchase of hybrid vehicles. Our Boston office could benefit from several incentives according to Senate Bill 1380, most notably a \$2,000 income tax deduction. Other states where our offices are located, such as Missouri, New York, and Ohio, offer \$1,500 to \$3,000 in tax incentives for the purchase of hybrid vehicles.

Parenthetical citation at end of sentence when source is not cited in sentence text

Federal

According to Angela Neville, in *Electric Vehicles: The Uncertain Road Ahead* (2010), in 2009 the federal government enacted a tax credit of \$2,500 to \$7,500 for individuals who buy a hybrid vehicle.

COSTS

Initial costs for a fleet of hybrids may seem high, but there are ways of reducing the cost of maintenance.

• Purchase

Hybrid vehicles are more expensive than conventional gasoline vehicles. The base model Tahoe starts at \$37,980, and the hybrid Tahoe retails at \$51,145 (Chevrolet, 2011b). Two of the least expensive hybrid sedans are the Toyota Prius (\$23,050) and Honda Fit Hybrid (17,00), which is going to be released in the United States this year (Dubois, 2010). The least expensive hybrid SUV is the Ford Escape (\$29,865). GMC now offers the Sierra Hybrid (\$38,710). The least expensive hybrid truck is the Chevrolet Silverado starting at 39,900 (HybridCars, 2011b).

• Maintenance

Most of the hybrid vehicles on the market come with standard three-year/36,000-mile warranties and eight-year/100,000-mile warranties on the batteries. Routine maintenance should probably be performed at a dealer, especially if the car is under warranty. In an article written by Erik Sofrge in *Popular Mechanics* (2010), a San Francisco cab company reduced its break pad changes from every 10,000 miles to 50,000 miles. Such a reduction could decrease the amount of time and money M-Global spends maintaining the company's fleet. Many companies are also developing faster methods to charge the batteries. Another option is to provide certification to the M-Global fleet mechanics so that they can perform maintenance on hybrid vehicles themselves. Because the market for hybrid vehicles has become so popular, technician certification programs are popping up all over the country in order to keep pace with the demand.

THE HYBRID MARKET

The hybrid market is expanding each year with new vehicles being added, such as the addition of trucks and minivans. In 2010 there were 23 hybrid vehicles on the market, and now in 2011 6 new models have been added, making the total number of hybrid vehicles 29 (Anderson, 2010). According to J. D. Power and Associates, by 2012, hybrids will account for 3.5 percent of the new-car market, with 44 different hybrids for sale (Automotive Editors, 2006). Honda and Toyota seem to always be competing for the newest hybrid technology. The Toyota Prius remains the top-selling hybrid vehicle, but Honda's CRZ will be the smallest hybrid, at 161 inches long and 15 inches shorter than the Toyota Prius, which increases the fuel efficiency and performance of the vehicle (HybridCars.com, 2011a).

THE OUTLOOK FOR HYBRID VEHICLES

Hybrid vehicles have a promising future, even as research continues on other technology.

• Near Future

With the rising gas prices and growing concern for the environment, hybrid vehicles are an excellent way to save money and contribute to the reduction of greenhouse gases. Although most experts agree that hybrid vehicles are only a temporary alternative to conventional internal-combustion automobiles, the next technology (fuel cell) is still many years from making itself into the mass market—particularly because fueling stations will have to become readily available to the general public.

• 2015 Predictions

Average gas prices are predicted to be around \$4.50/gallon in 2015. There may be higher gas prices between now and then, but the investment to find additional reserves will be justified by the oil companies, and prices could fluctuate (Belzowski, 2006). This rise in gas prices could affect the way that Americans purchase their vehicles, and it is possible that hybrids could become the predominant vehicle on the road.

• 2020 Predictions

Average gas prices are predicted to be around \$6.00/gallon in 2020. It is estimated that replacement energy will start and gas prices will drop (Belzowski, 2006). Many experts believe that this replacement energy will be in the form of hydrogen fuel cells, but others disagree because of the challenges that fuel cell technology must overcome. The biggest challenges are a low-cost source of hydrogen and a hydrogen infrastructure (Belzowski, 2006). The possibility remains that hybrid vehicles will still be the predominant vehicle on the road, even after 2020.

CONCLUSION

Initially, the purchase of hybrid vehicles would be more expensive, but we have a good chance of making up the initial expense in saved fuel costs and tax savings.

Because the makers of hybrid vehicles now offer sedans, SUVs, and trucks, we should be able to replace every vehicle in our fleet with a hybrid version in the same class. Furthermore, we can consider buying our vehicles using our discount arrangement with General Motors, as it now offers hybrid versions of the Malibu, Tahoe, and Silverado.

M-Global is well known for its environmental services, and having part or all of our fleet go green could help further emphasize this positive image in the minds of our customers.

BIBLIOGRAPHY

- Anderson, J. (2010, September). A green car for every driver. *Kiplinger's Personal Finance*. Retrieved from <http://web.ebscohost.com>.
- Automotive Editors. (2006, February). Comparison—Hybrids at the crossroads. *Popular Mechanics*, 183(2). Retrieved from <http://www.popularmechanics.com>.
- Belzowski, B. M. (2006, October). *Powertrain strategies for the 21st century*. Ann Arbor: University of Michigan Transportation Research Institute. Retrieved from <http://www.osat.umich.edu/research/powertrain/NAPowertrainReportFinal1.pdf>.
- Bickerstaffe, S. (2007). Cutting the cost of hybrids. *Automotive Engineer*, 32(5), 34.
- Bullis, K. (2007). Electric cars 2.0. *Technology Review*, 110(5), 100–101.

Clearly indicates how reader should use this research.

Bibliography list starts on new page because will not fit at end of last page of report

Do not use articles (a, an, or the) when alphabetizing list of references

- Chevrolet. (2011a). *2011 Malibu*. Retrieved from <http://www.chevrolet.com/malibu/features-specs>.
- Chevrolet. (2011b). *2011 Tahoe*. Retrieved from <http://www.chevrolet.com/tahoe-family>.
- Dubois, N. (2010, December 22). Cheapest hybrid cars. Retrieved from http://www.ehow.com/list_7677116_cheapest-hybrid-cars.html.
- Dworetzky, T., Hart, H., John, T., Labrecque, J., & Thill, S. (2009, December). Gentlemen, don't start your engines. *Discover* 30(5), 39. Retrieved from <http://web.ebscohost.com>.
- Global Business Briefs. (2007, June 10). *Wall Street Journal*, p. 11A
- HybridCars.com. (2006). *Electromagnetic fields in hybrids*. Retrieved from <http://www.hybridcars.com/safety/electromagnetic-fields-in-hybrids.html>.
- HybridCars.com. (2011a). *Hybrid EMF risk still uncertain*. Retrieved from <http://www.hybridcars.com/safety/hybrid-emf-risk-still-uncertain.html>.
- HybridCars.com. (2011b). *Top hybrid cars: A complete guide*. Retrieved from <http://www.hybridcars.com/top-hybrid-cars-list>.
- HybridCar.org. (n.d.). *Hybrid car emissions*. Retrieved from <http://www.hybrid-car.org/hybrid-car-facts.html>.
- In search of the perfect battery. (2008, March 8). *The Economist*, 386(8570), 22–24.
- Internal Revenue Service. (2007, November 8). *Summary of the credit for qualified hybrid vehicles*. Retrieved from <http://www.irs.gov/newsroom/article/0,,id=157557,00.html>.
- Missouri Senate Bill 163. (2009). *Missouri Senate*. Retrieved April 2, 2011 from <http://www.senate.mo.gov/09info/pdf-bill/intro/SB163.pdf>.
- Motivalli, J. (2008, April 27). Fear, but few facts, on hybrid risk. *New York Times*. Retrieved from <http://www.nytimes.com>.
- MSN Autos Editors. (n.d.). Most popular hybrids on MSN. Retrieved from <http://editorial.autos.msn.com>.
- Neville, A. (2010, March). Electric vehicles: The uncertain road ahead [electronic version]. *Power*. Retrieved <http://web.ebscohost.com>.
- Northeast Advanced Vehicle Consortium. (n.d.). *Heavy Duty hybrid vehicle testing: Particulate matter (PM) emissions*. Retrieved from <http://www.navc.org/HDPM.html>.
- Richard, M. G. (2008, April 15). *Plug-in hybrid cars: Chart of CO2 emissions ranked by power source*. Retrieved from <http://www.treehugger.com/files/2008/04/plug-in-hybrid-cars-co2-emissions-electricity-energy.php>.
- Sofrge, E. (2010, June). The electric plug-in acid test. *Popular Mechanics*, 187(10). Retrieved from <http://web.ebscohost.com>.
- Toyota Prius. (2011). *2011 Toyota Prius*. Retrieved from <http://www.toyota.com/prius-hybrid/specs.html>.
- U.S. Environmental Protection Agency. (n.d.). *How hybrids work*. Retrieved from <http://www.fueleconomy.gov/feg/hybridtech.shtml>.
- U.S. Environmental Protection Agency. (2008a, January 16). *2008 Fuel Economy Guide*. Retrieved from <http://www.fueleconomy.gov/feg/FEG2008.pdf>.
- U.S. Environmental Protection Agency. (2008b, January 17). *New energy tax credits for hybrids*. Retrieved from http://www.fueleconomy.gov/feg/tax_hybrid.shtml.
- U.S. Environmental Protection Agency. (2010, November 3). *2011 Fuel Economy Guide*. Retrieved from <http://www.epa.gov/fueleconomy/overall-high.htm>.
- U.S. Environmental Protection Agency. (2011, March 30). *Green Vehicle Guide*. Retrieved from <http://www.epa.gov/greenvehicles/Index.do>.
- Weisenfelder, J. (n.d.). *Top 10 2008 hybrids*. Retrieved from http://autos.yahoo.com/articles/autos_content_landing_pages/469/top-10-2.